

Model Specification Sensitivity – Gait Speed

Reference transform: identity | Random slope: FALSE

Generated: 2026-06-28 11:37

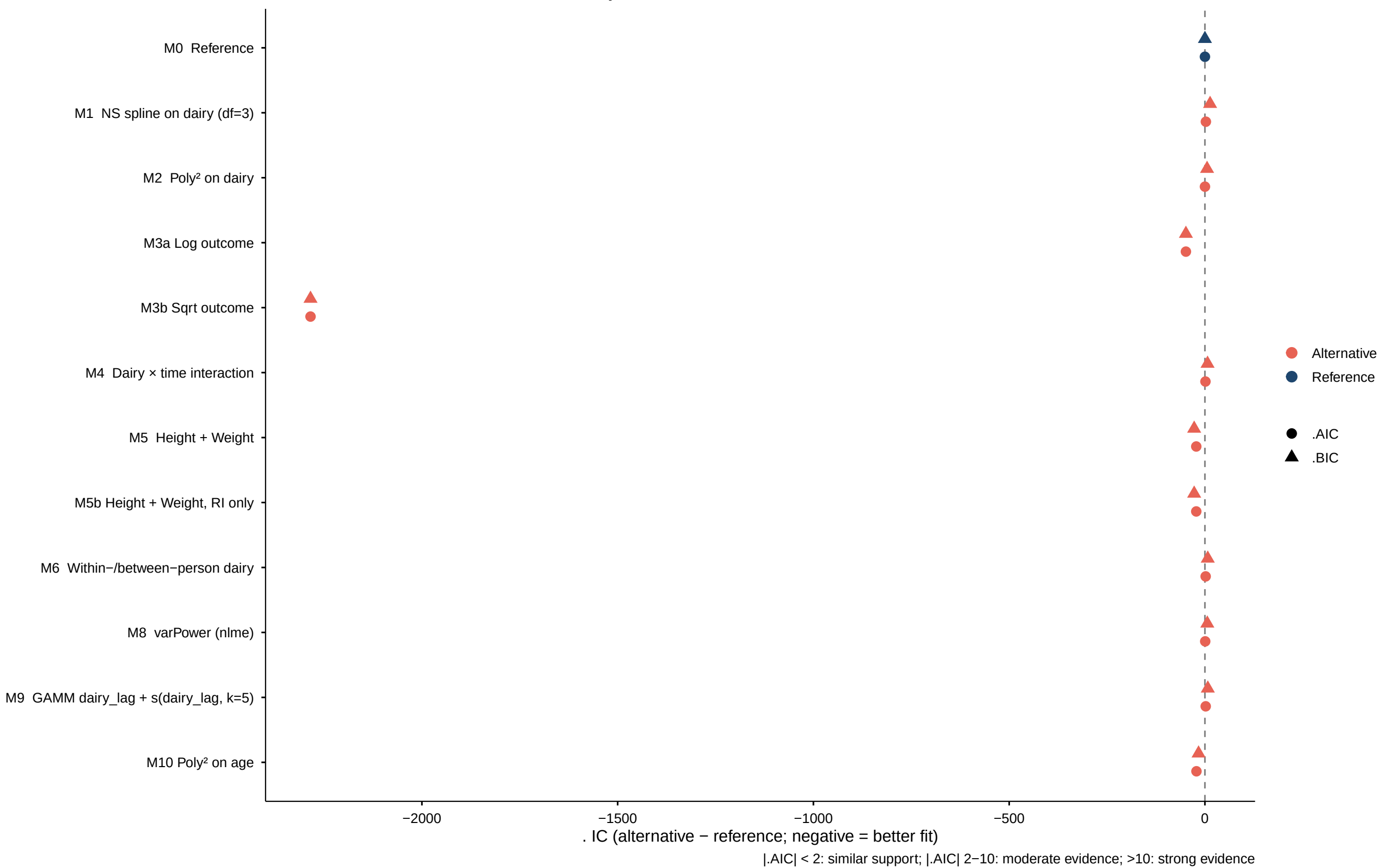
NOTE: AIC / BIC computed on imputation 1 only.

AIC / BIC Comparison

Gait speed

model	AIC	BIC	delta_AIC	delta_BIC	df_model	converged
M0 Reference	-1269.68	-1183.63	0.00	0.00	16	TRUE
M1 NS spline on dairy (df=3)	-1267.26	-1170.46	2.42	13.18	18	TRUE
M2 Poly ² on dairy	-1269.60	-1178.18	0.08	5.46	17	TRUE
M3a Log outcome	-1318.26	-1232.22	-48.58	-48.58	16	TRUE
M3b Sqrt outcome	-3554.14	-3468.10	-2284.47	-2284.47	16	TRUE
M4 Dairy × time interaction	-1268.36	-1176.94	1.32	6.70	17	TRUE
M5 Height + Weight	-1291.81	-1211.14	-22.13	-27.51	15	TRUE
M5b Height + Weight, RI only	-1291.81	-1211.14	-22.13	-27.51	15	TRUE
M6 Within-/between-person dairy	-1267.83	-1176.41	1.84	7.22	17	TRUE
M7 With random slope	NA	NA	NA	NA	NA	FALSE
M8 varPower (nlme)	-1268.96	-1177.54	0.72	6.10	17	TRUE
M9 GAMM dairy_lag + s(dairy_lag, k=5)	-1267.68	-1176.26	2.00	7.38	17	TRUE
M10 Poly ² on age	-1291.32	-1199.90	-21.64	-16.26	17	TRUE

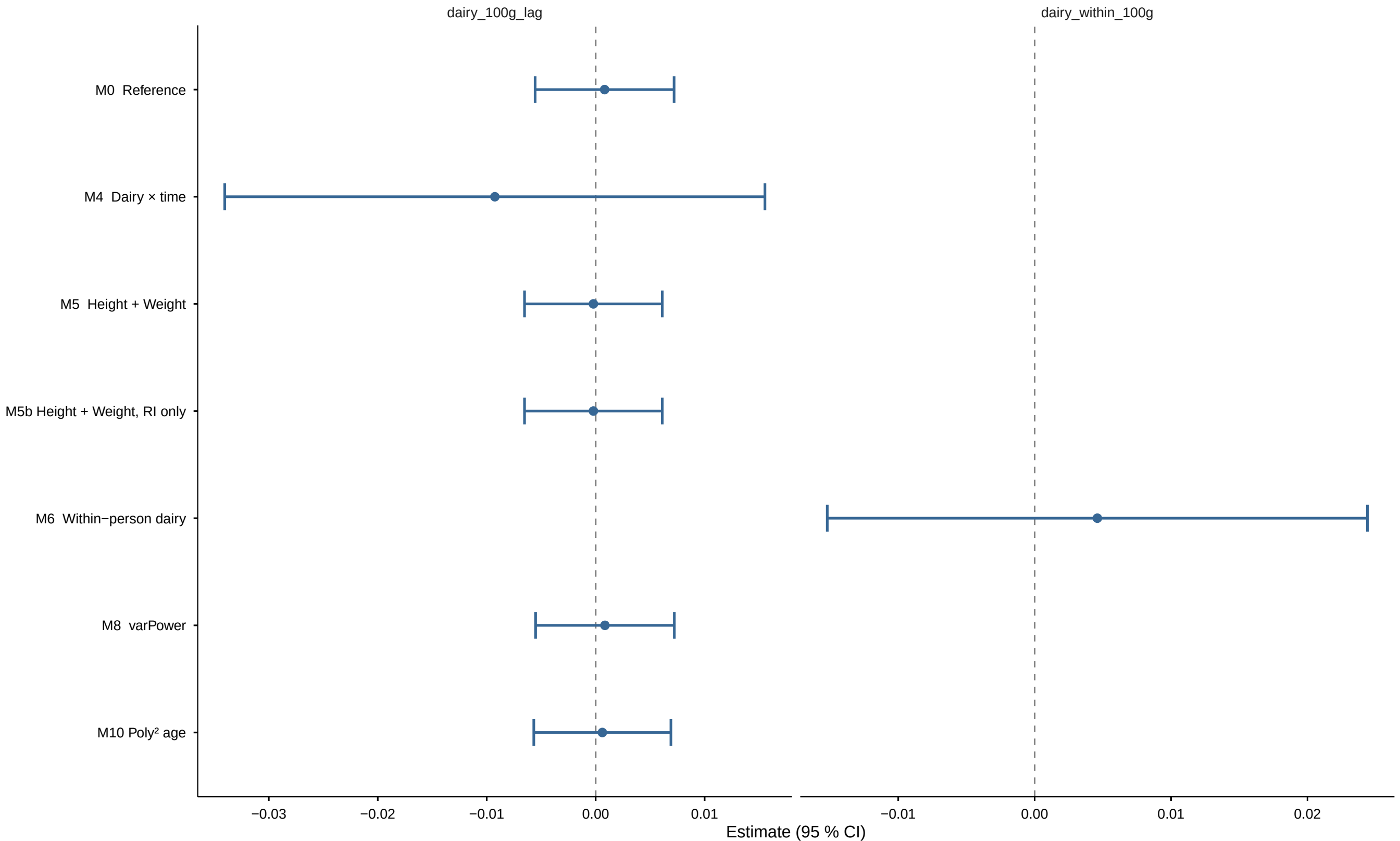
.AIC / .BIC relative to reference – Gait speed



Dairy Coefficient Comparison

Gait speed

Dairy (lagged) estimate (95 % CI) across specifications – Gait speed



Single imputation (imp 1); models using spline / poly dairy excluded

Assumption Checks

Gait speed

Assumption Checks – Notes

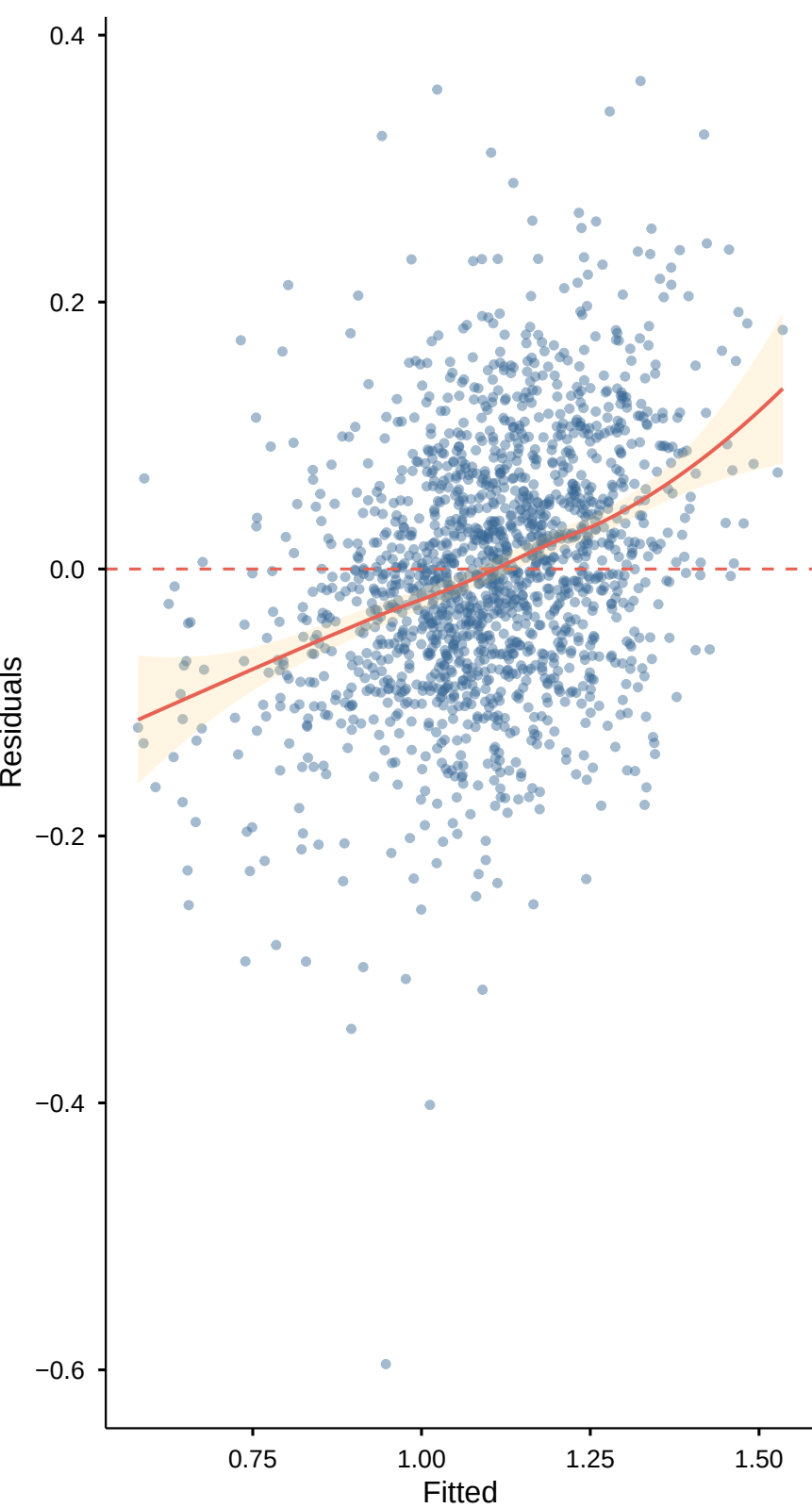
Each model: (A) Residuals vs Fitted, (B) Scale–Location, (C) Q–Q

M8 varPower (nlme): raw residuals are shown first, then Pearson / normalized residuals. Normalized residuals should be homoscedastic if the variance structure is correctly specified.

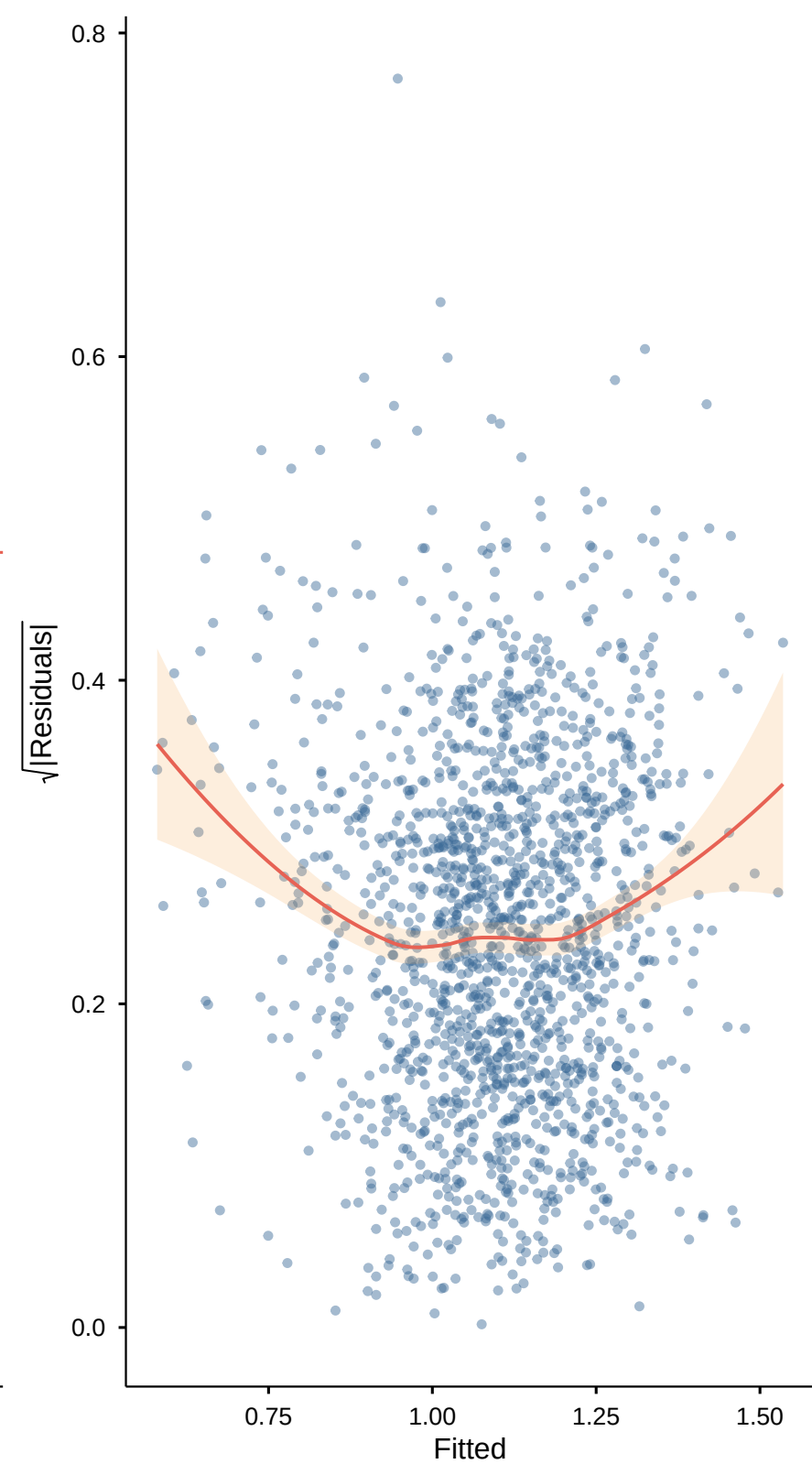
M9 GAMM: standard diagnostics use the lme component (conditional residuals at the subject level, comparable to lmer residuals). mgcv::gam.check() is appended as an additional GAMM–specific check.

Model Assumption Checks – M0 Reference

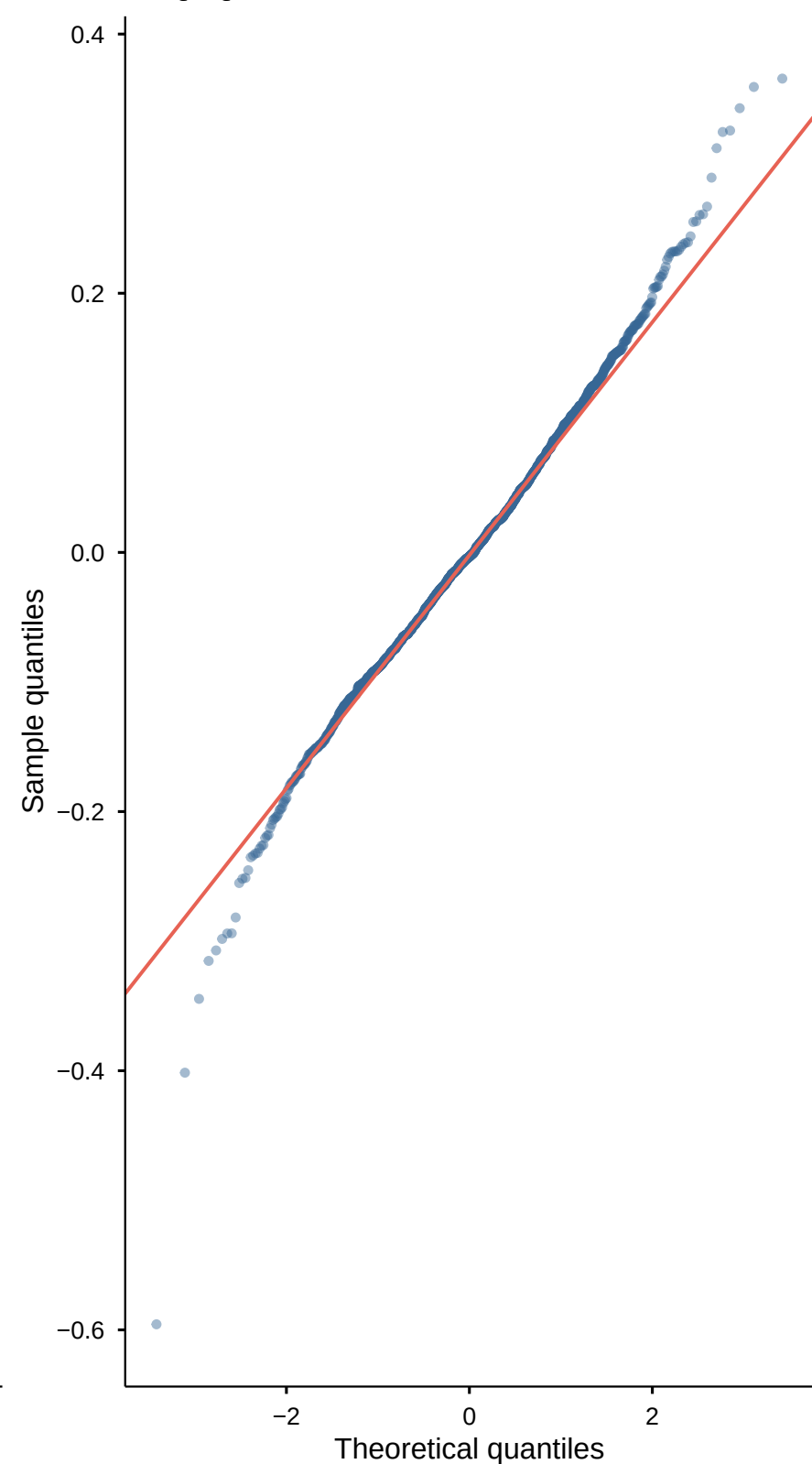
A Residuals vs Fitted



B Scale-Location

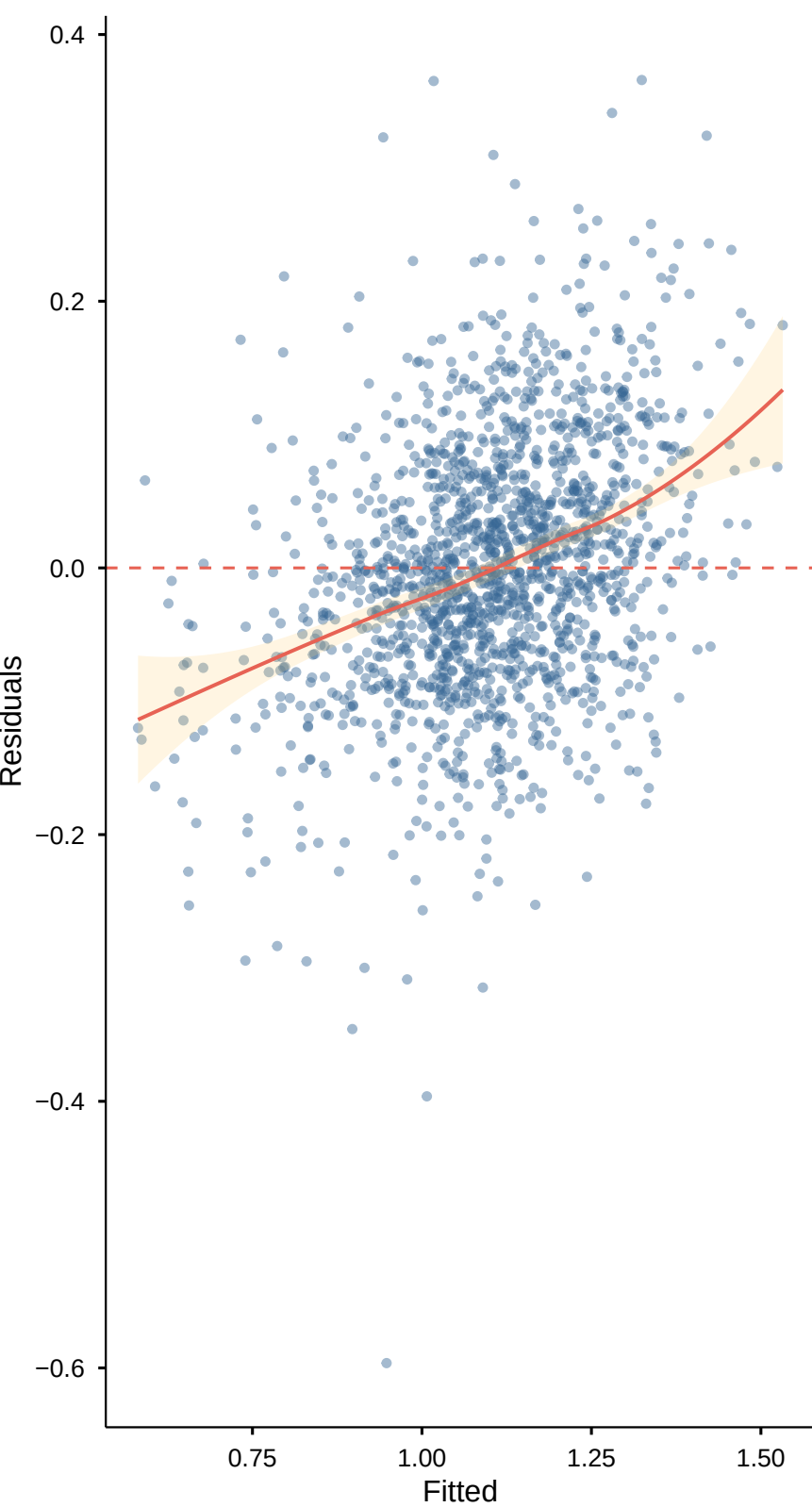


C Q-Q

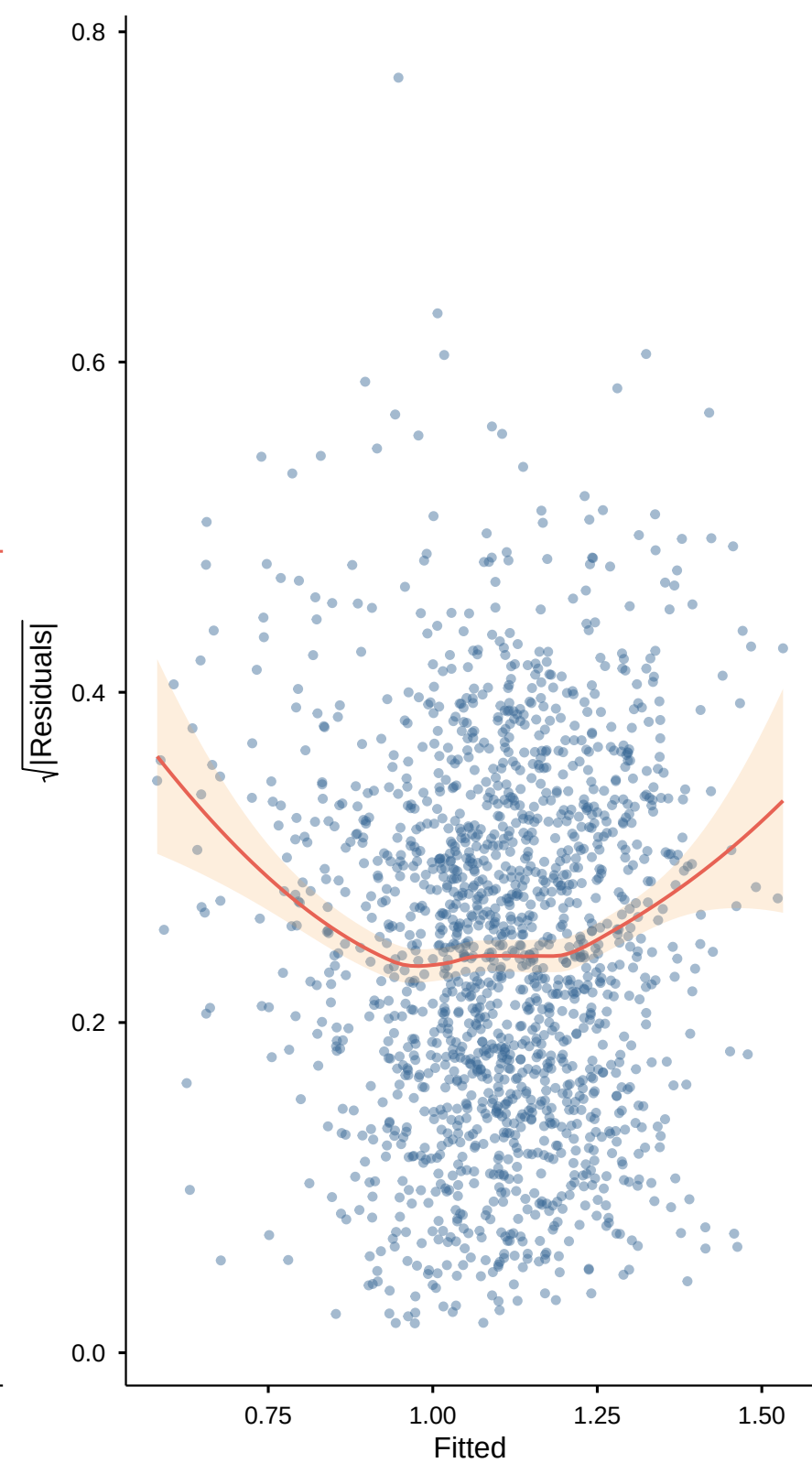


Model Assumption Checks – M1 NS spline dairy

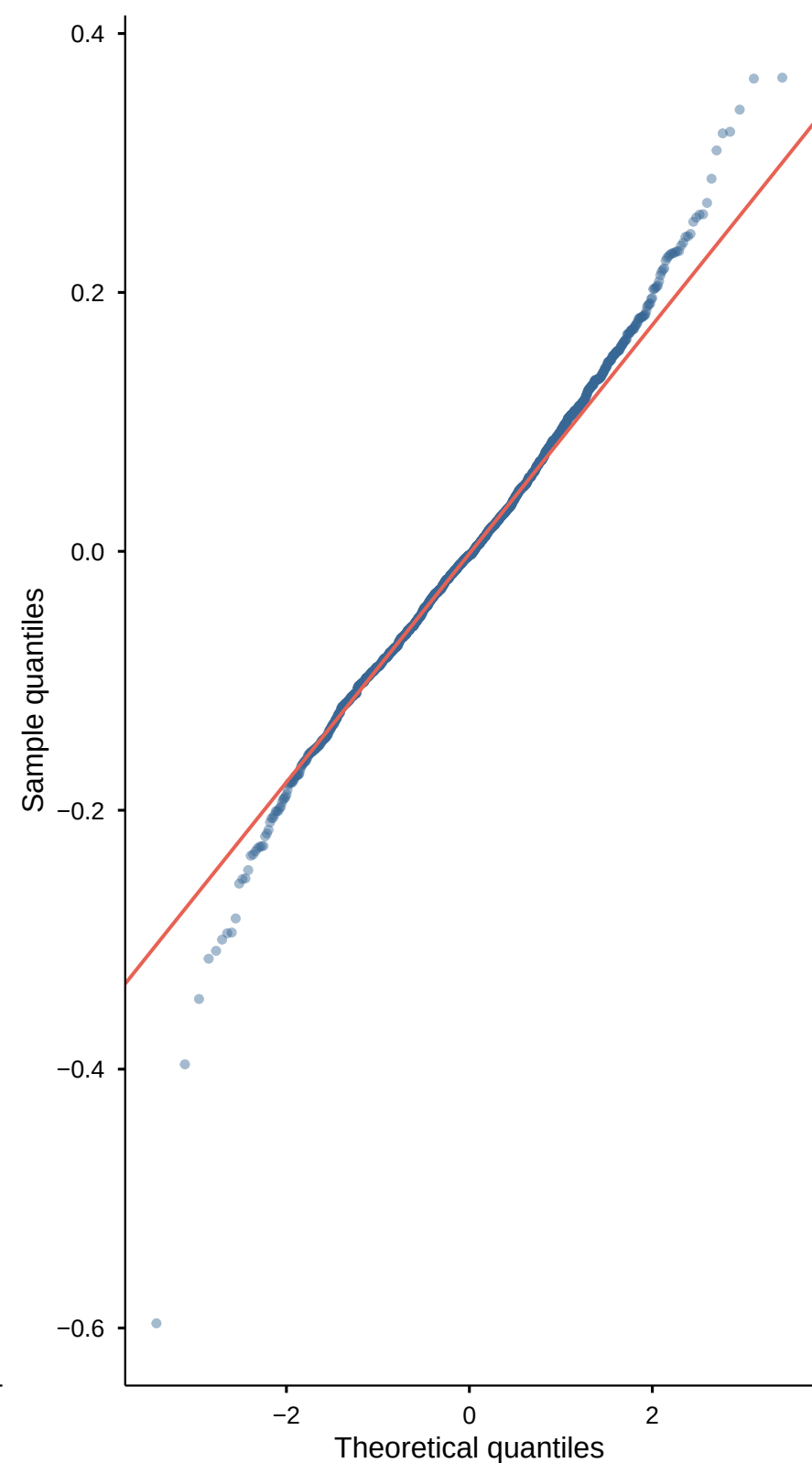
A Residuals vs Fitted



B Scale-Location

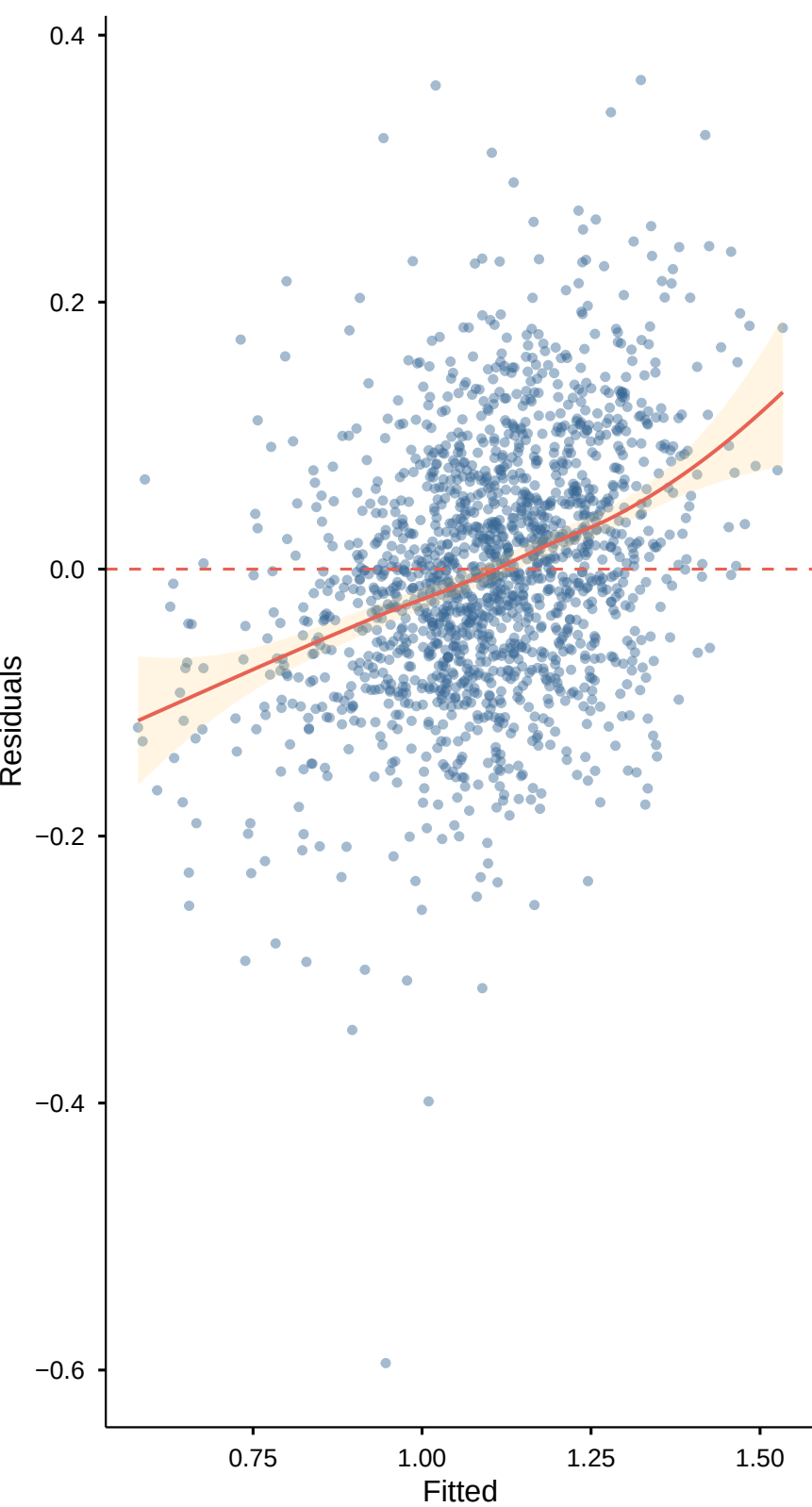


C Q-Q

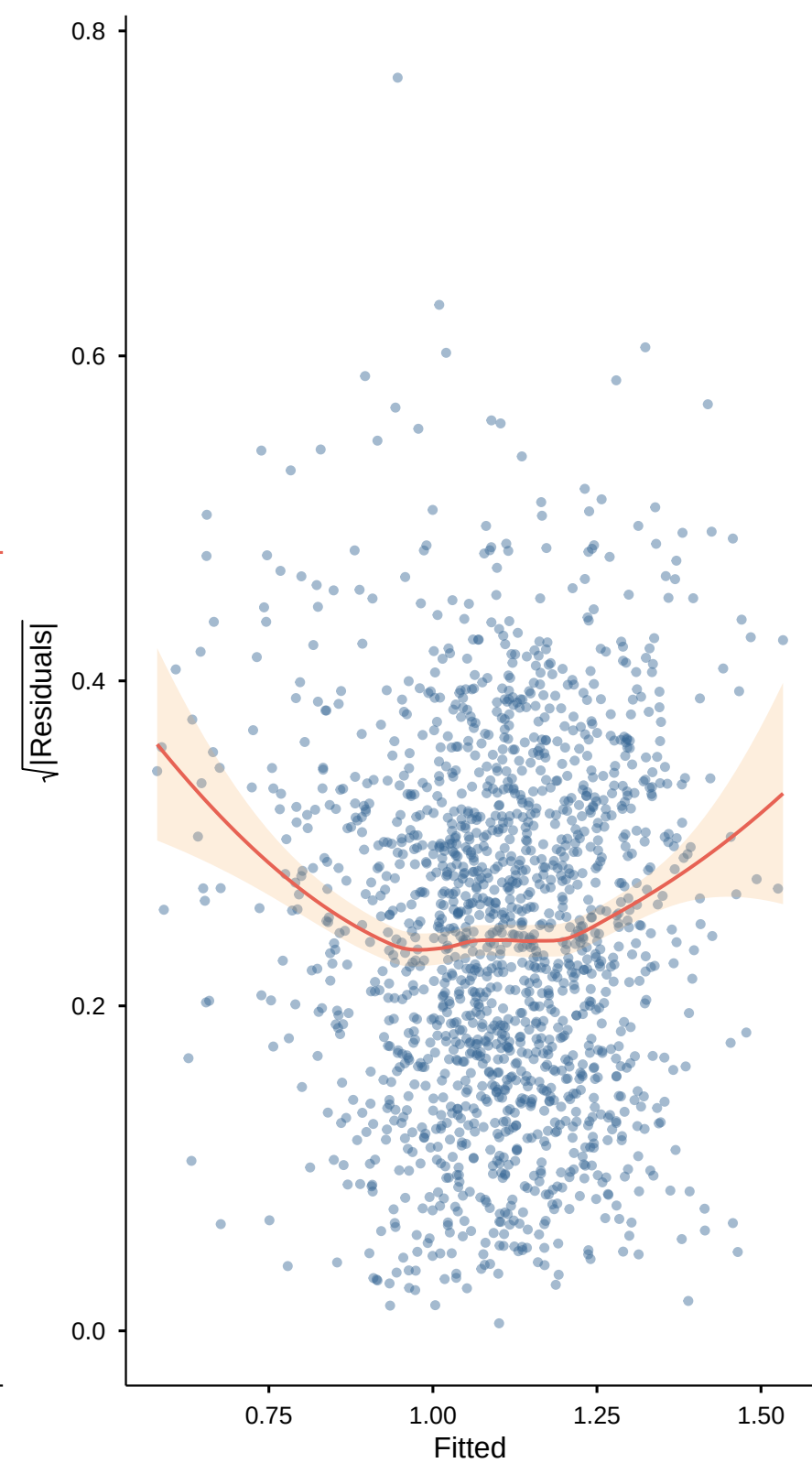


Model Assumption Checks – M2 Poly² dairy

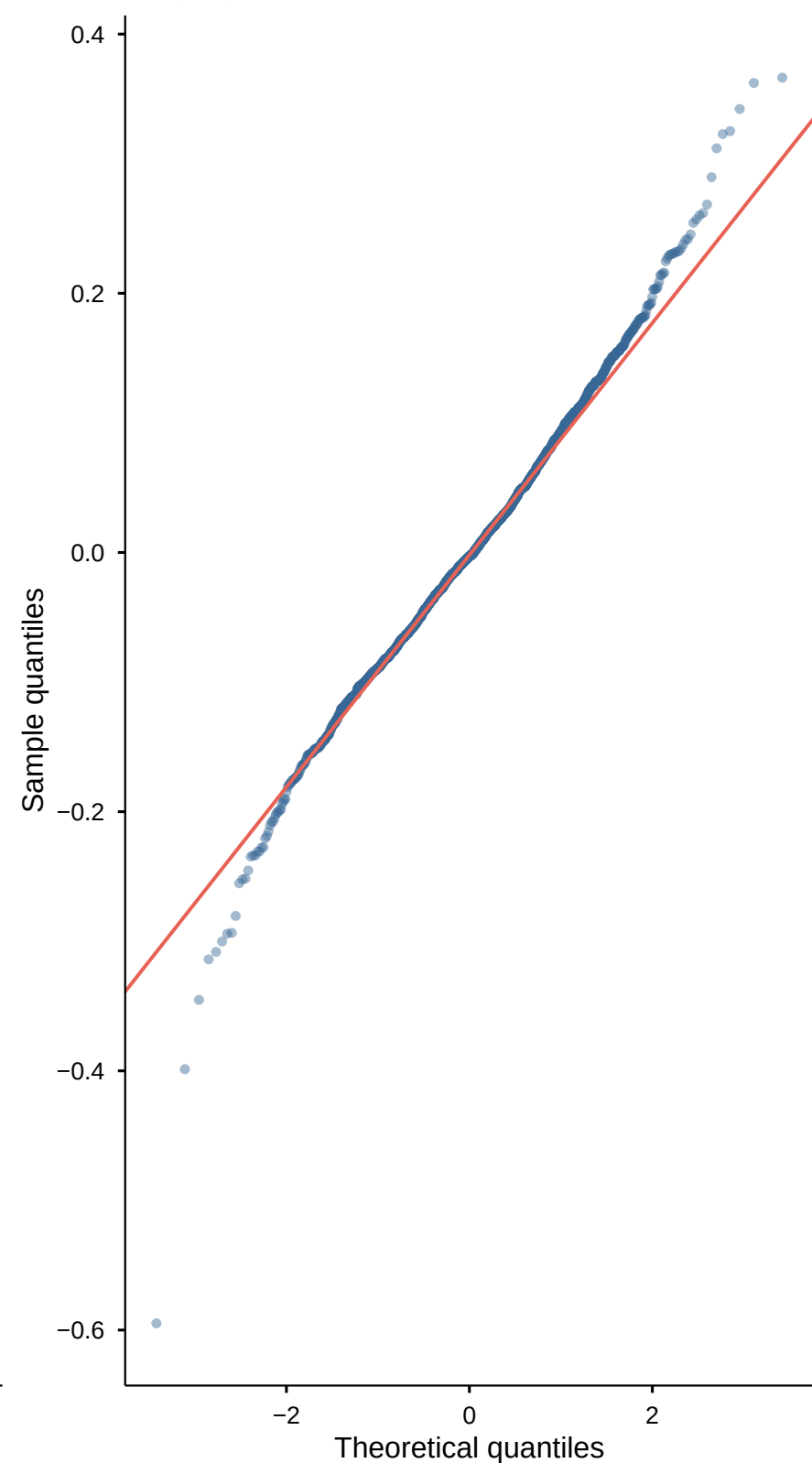
A Residuals vs Fitted



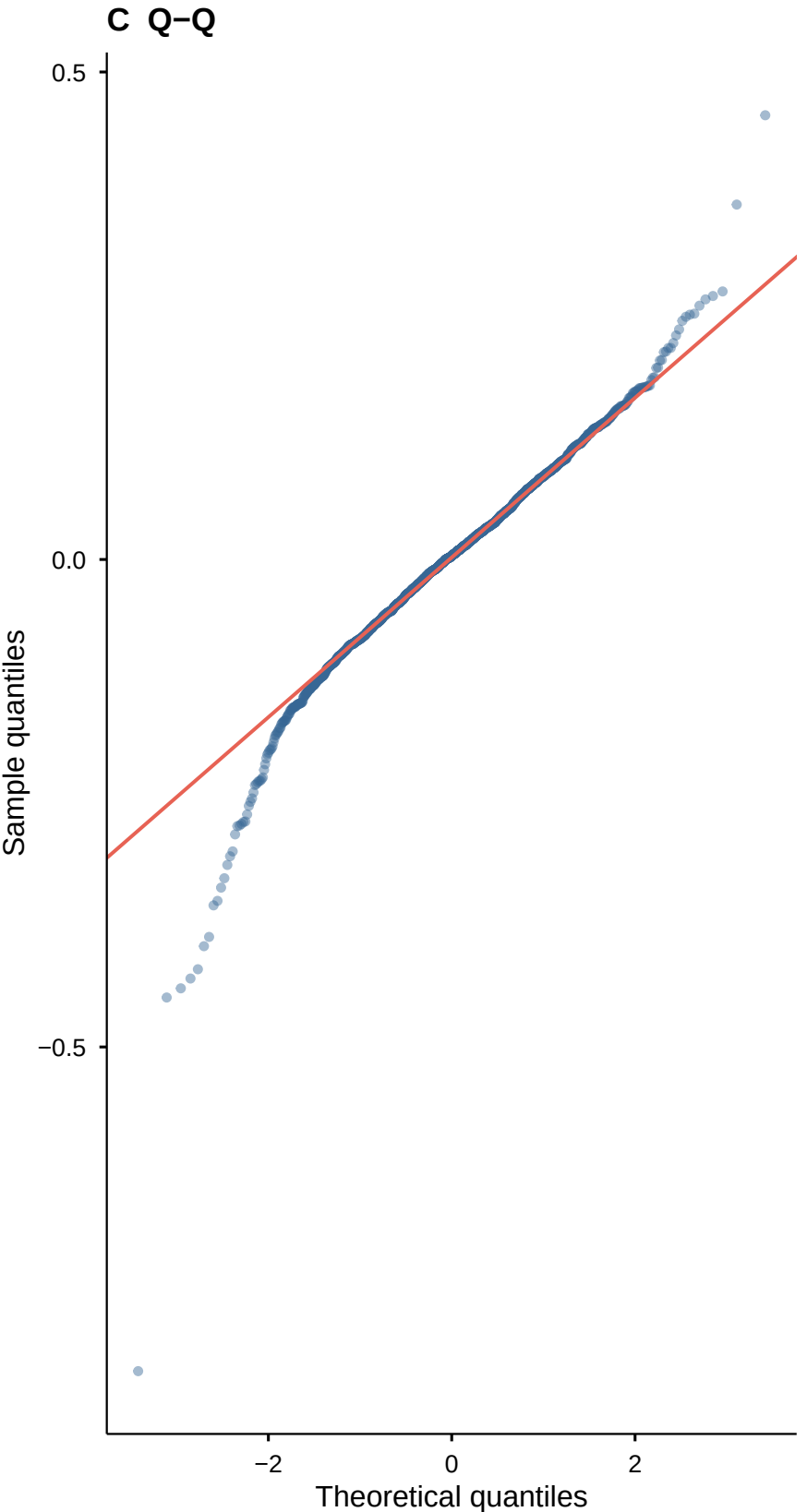
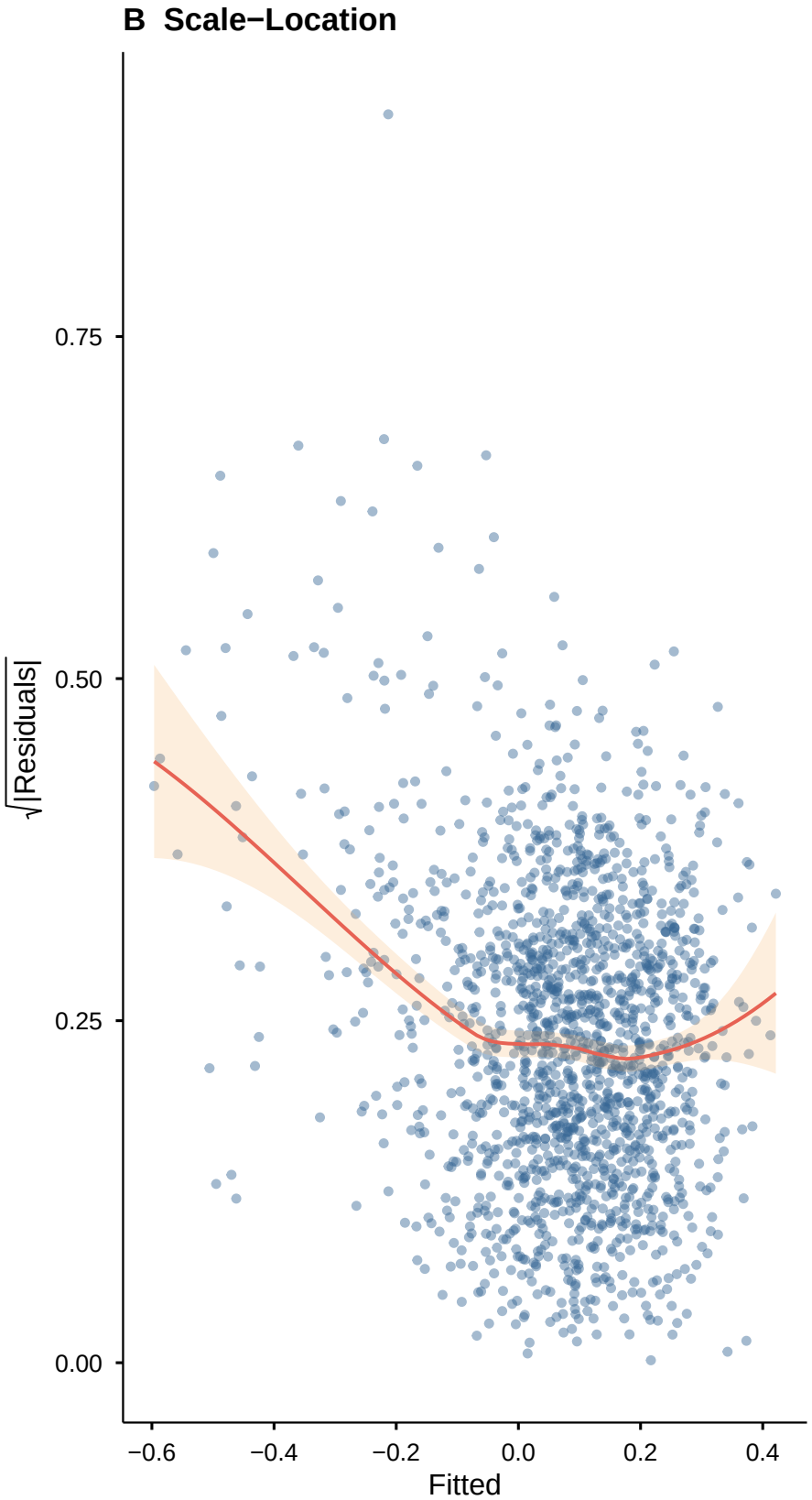
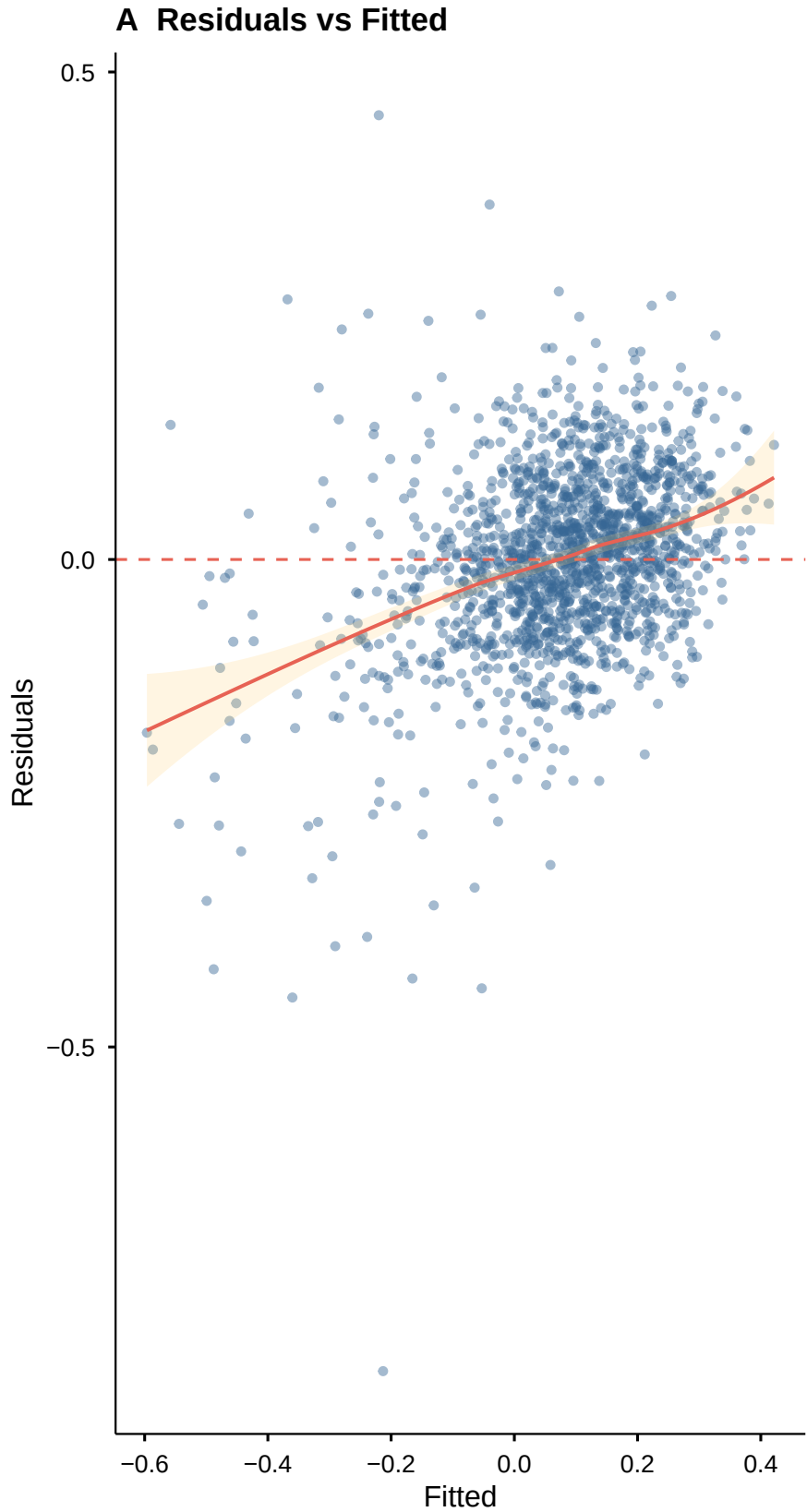
B Scale-Location



C Q-Q

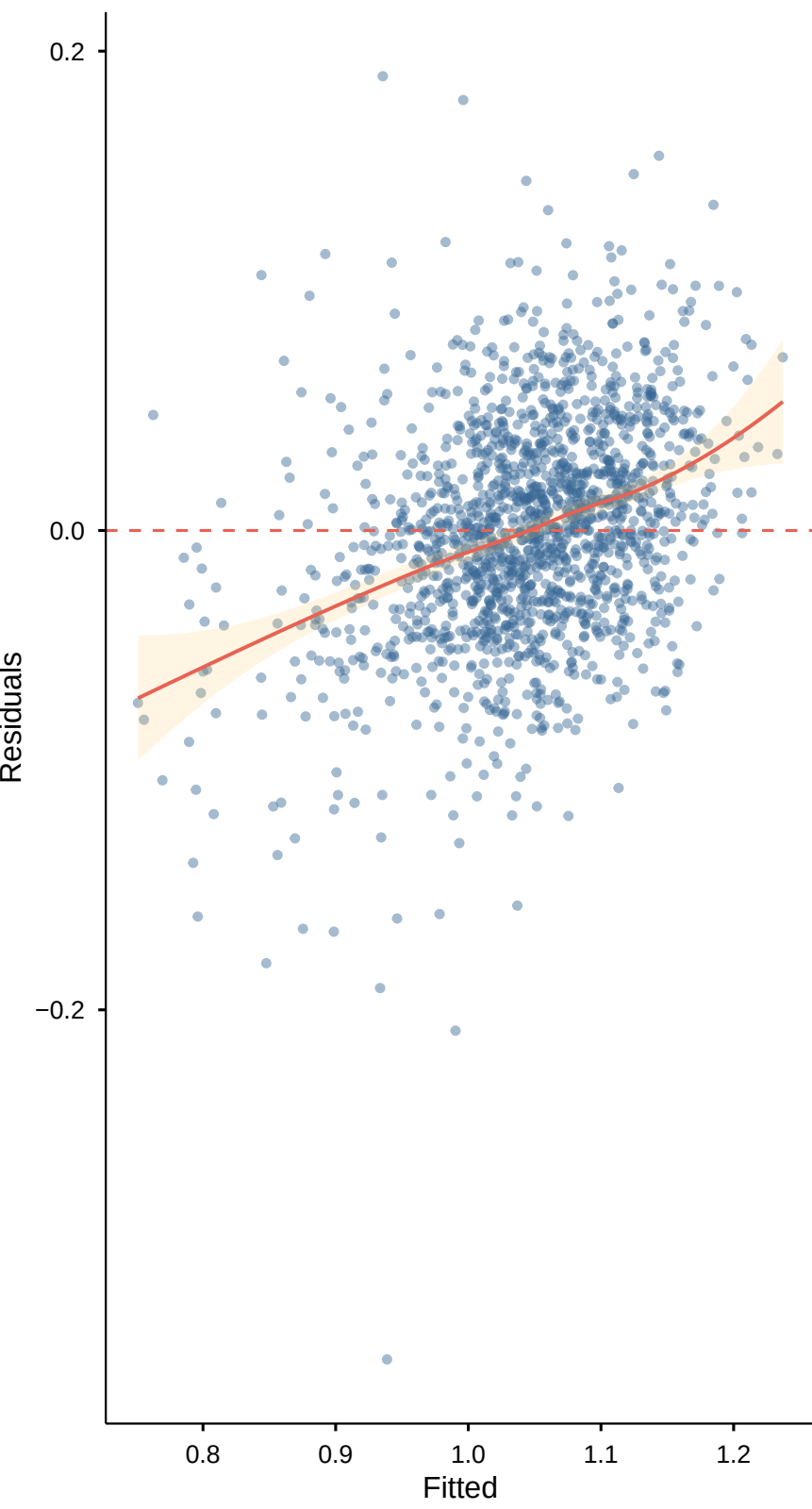


Model Assumption Checks – M3a Log outcome

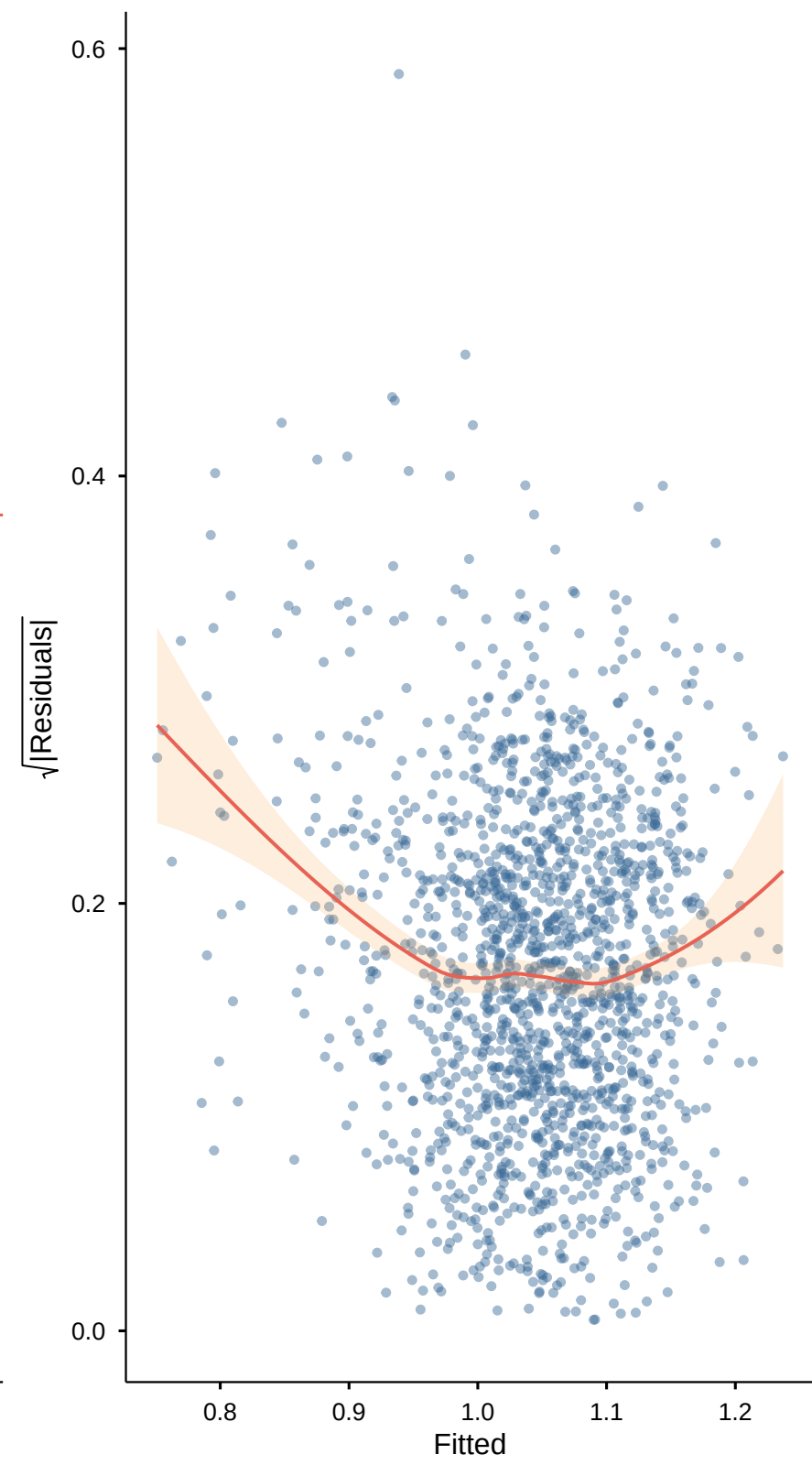


Model Assumption Checks – M3b Sqrt outcome

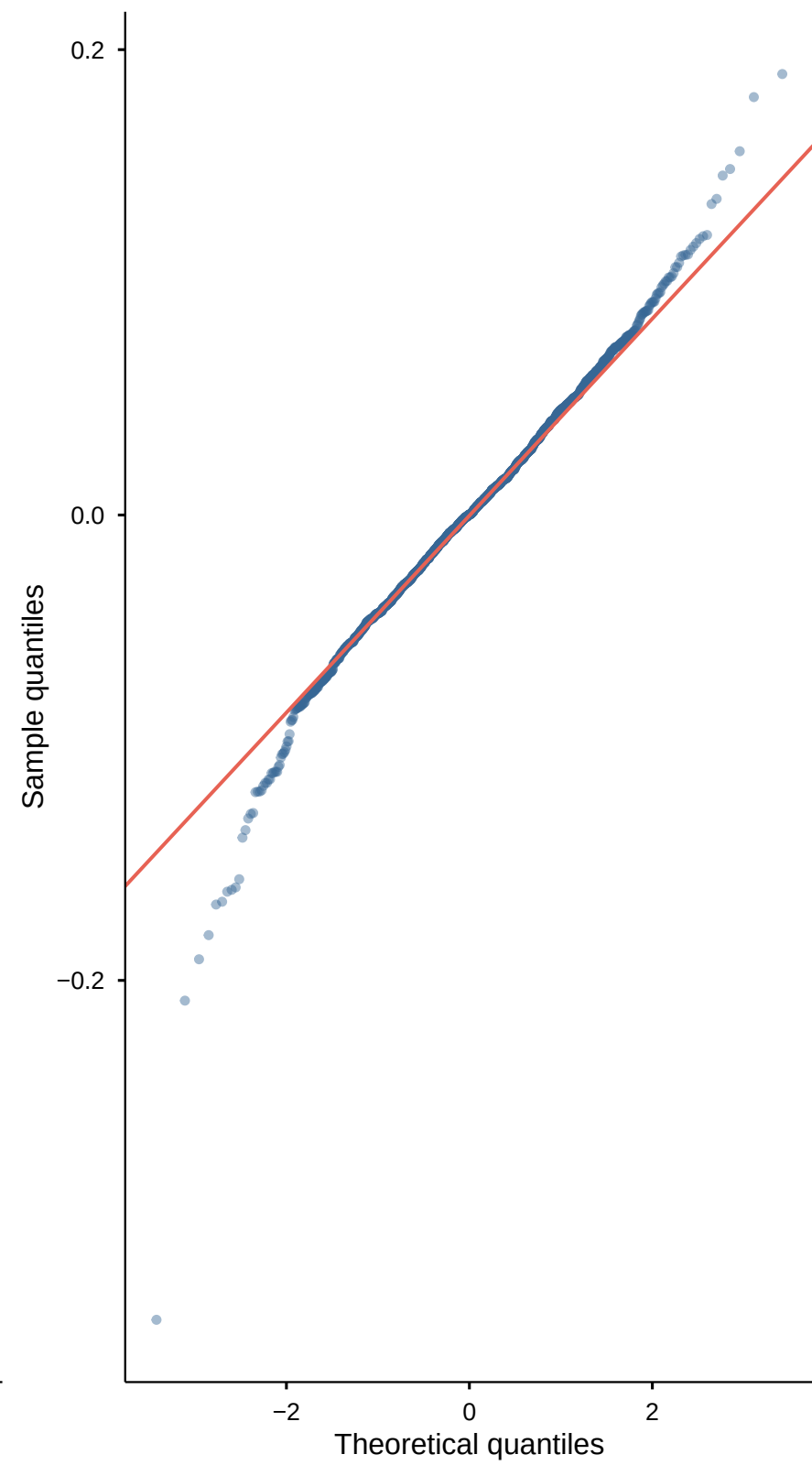
A Residuals vs Fitted



B Scale-Location

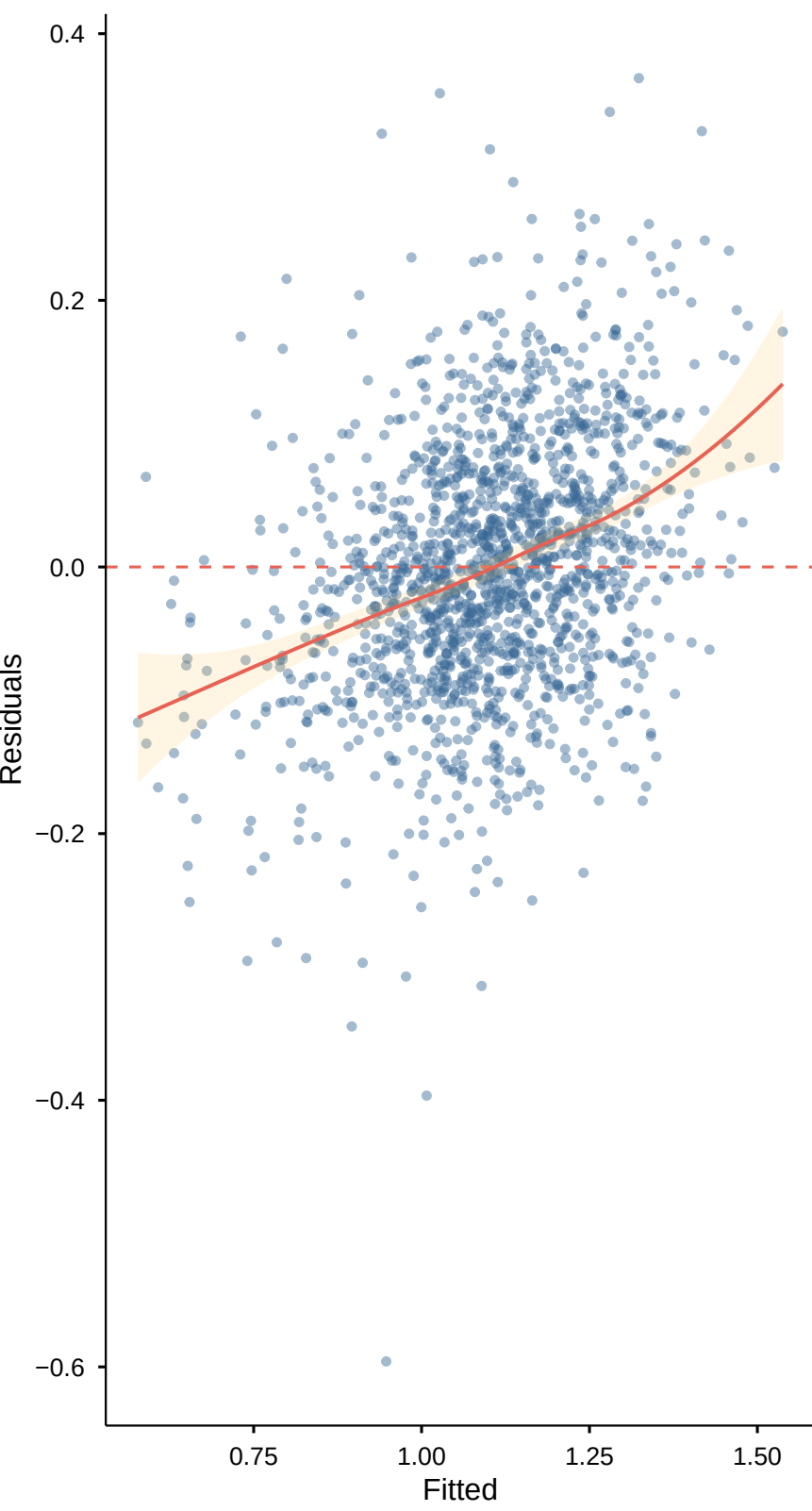


C Q-Q

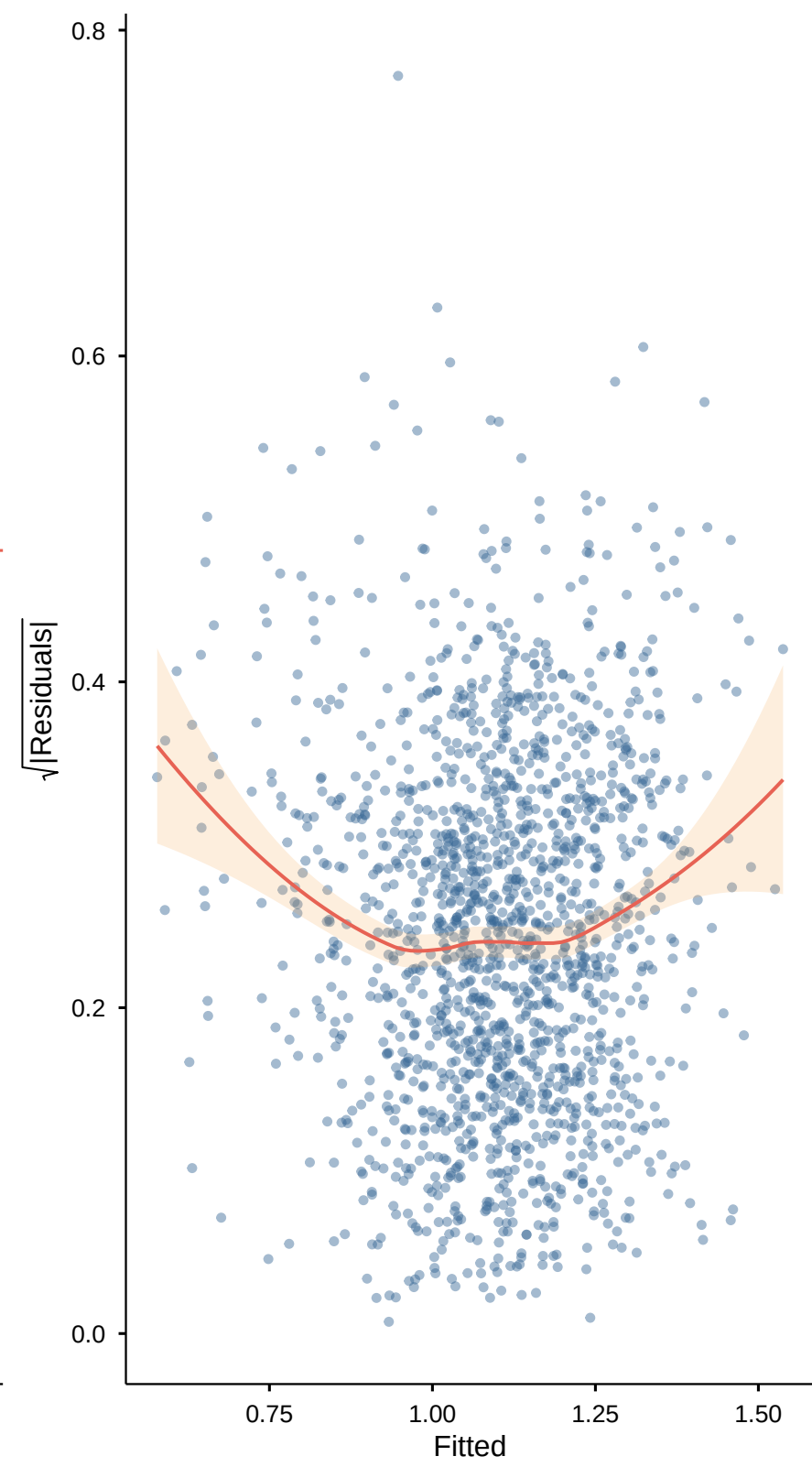


Model Assumption Checks – M4 Dairy × time

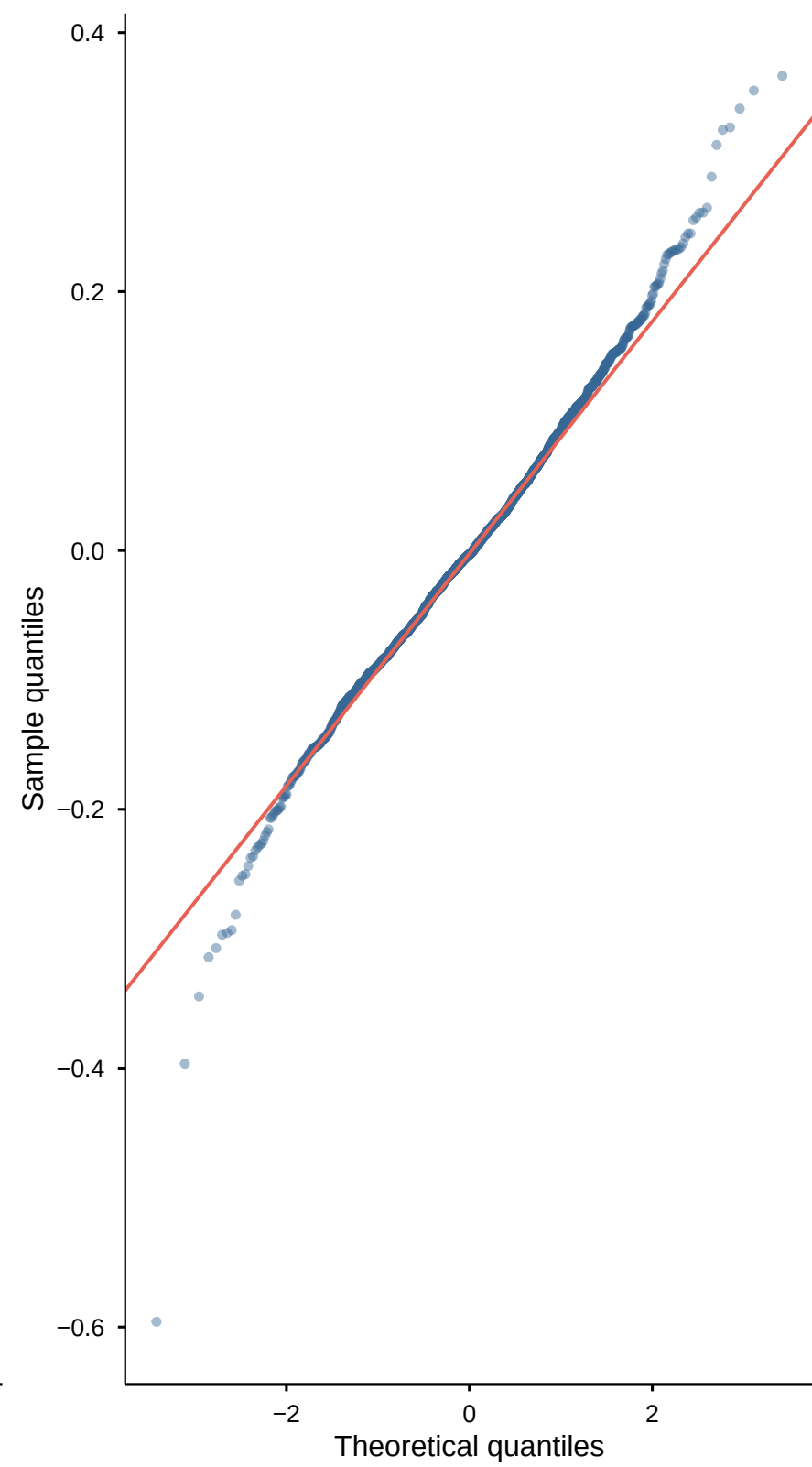
A Residuals vs Fitted



B Scale-Location

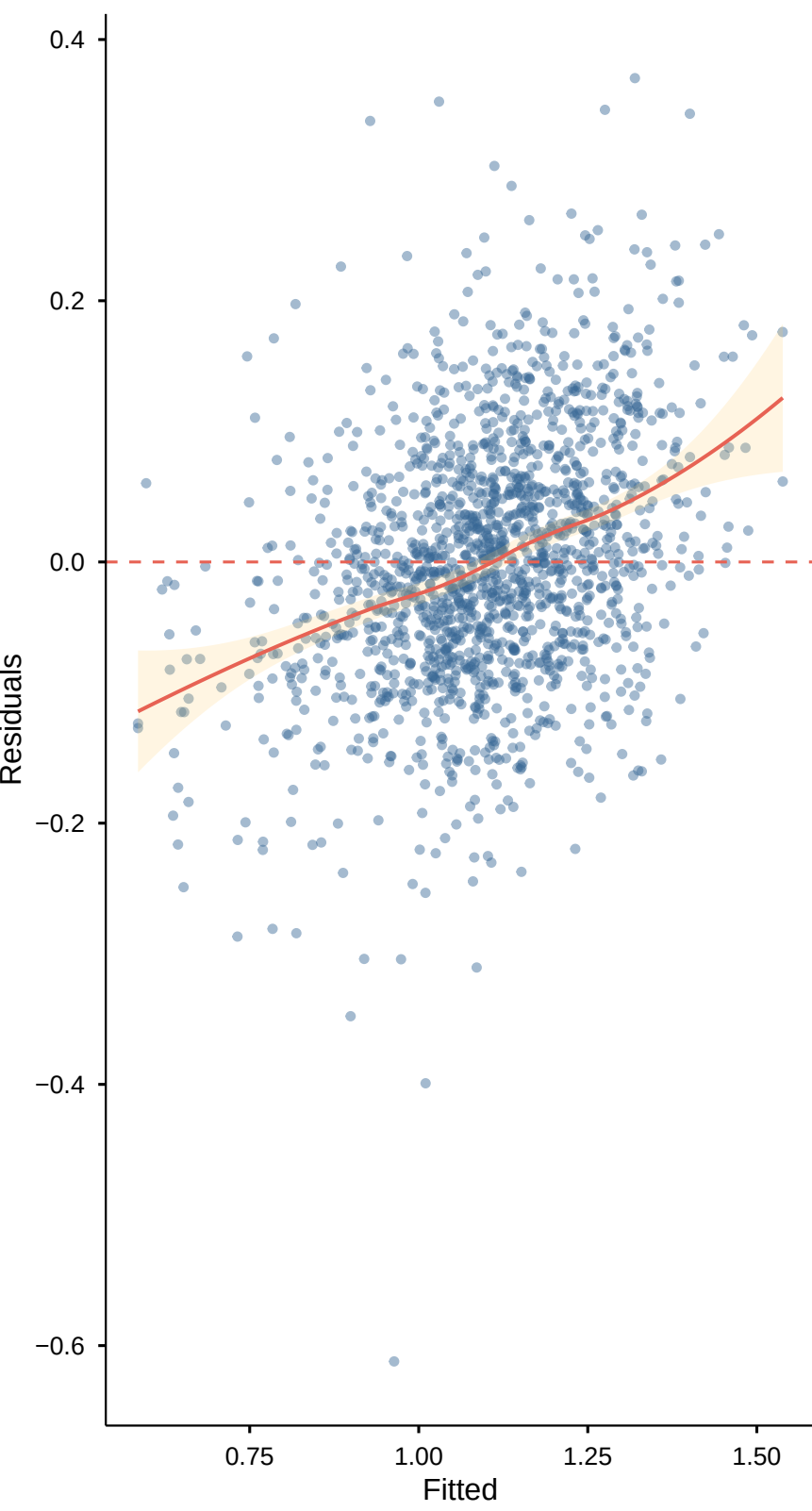


C Q-Q

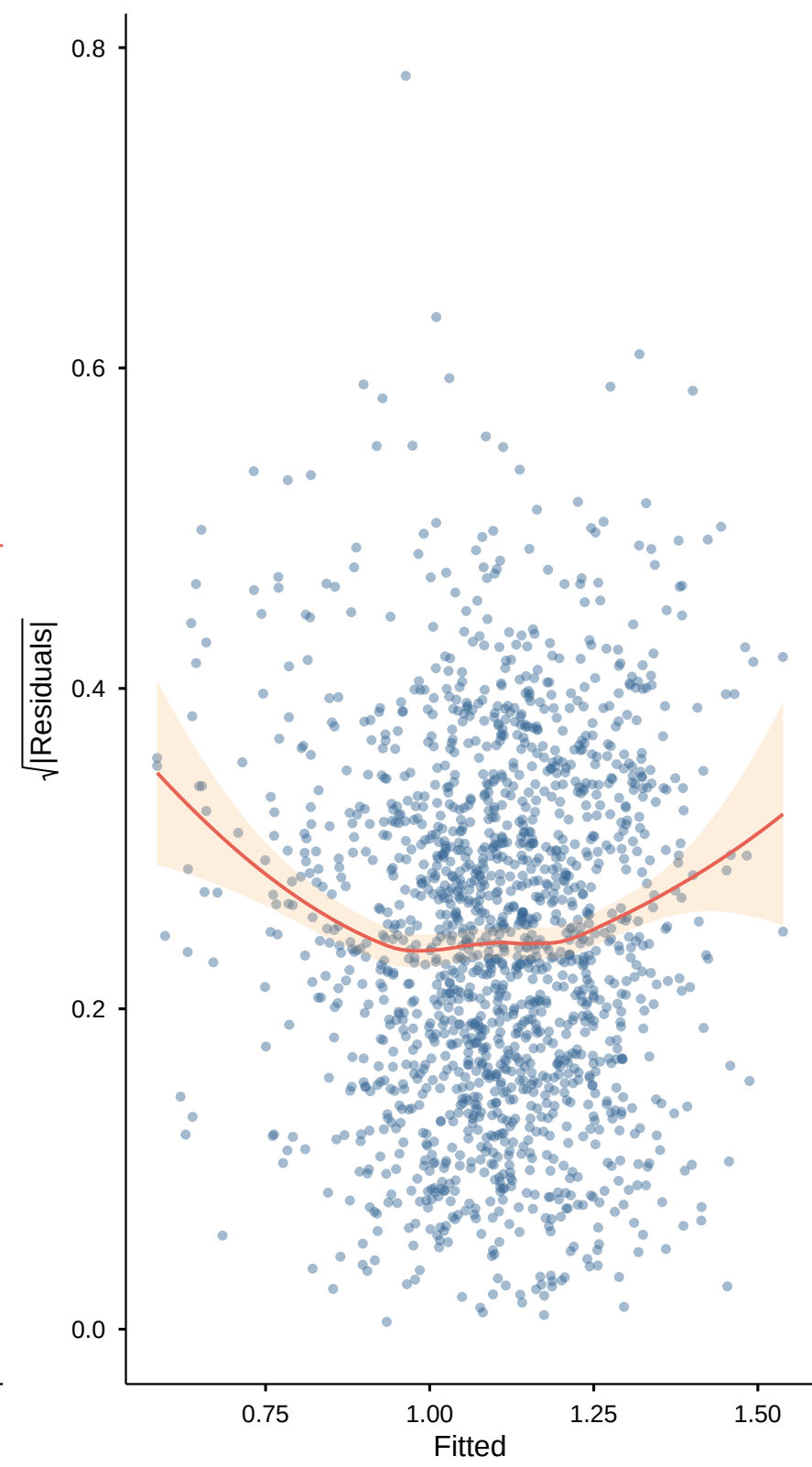


Model Assumption Checks – M5 Height + Weight

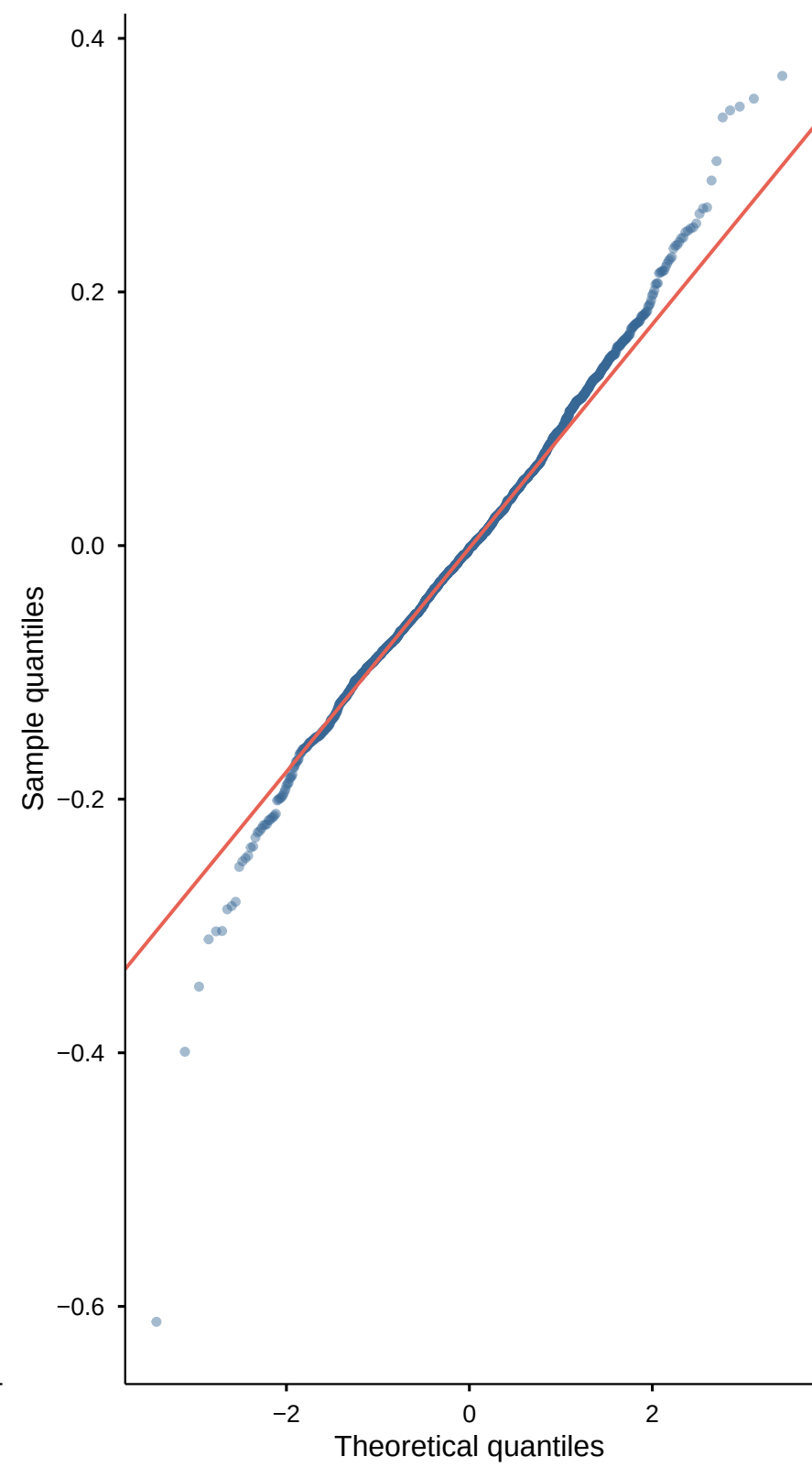
A Residuals vs Fitted



B Scale-Location

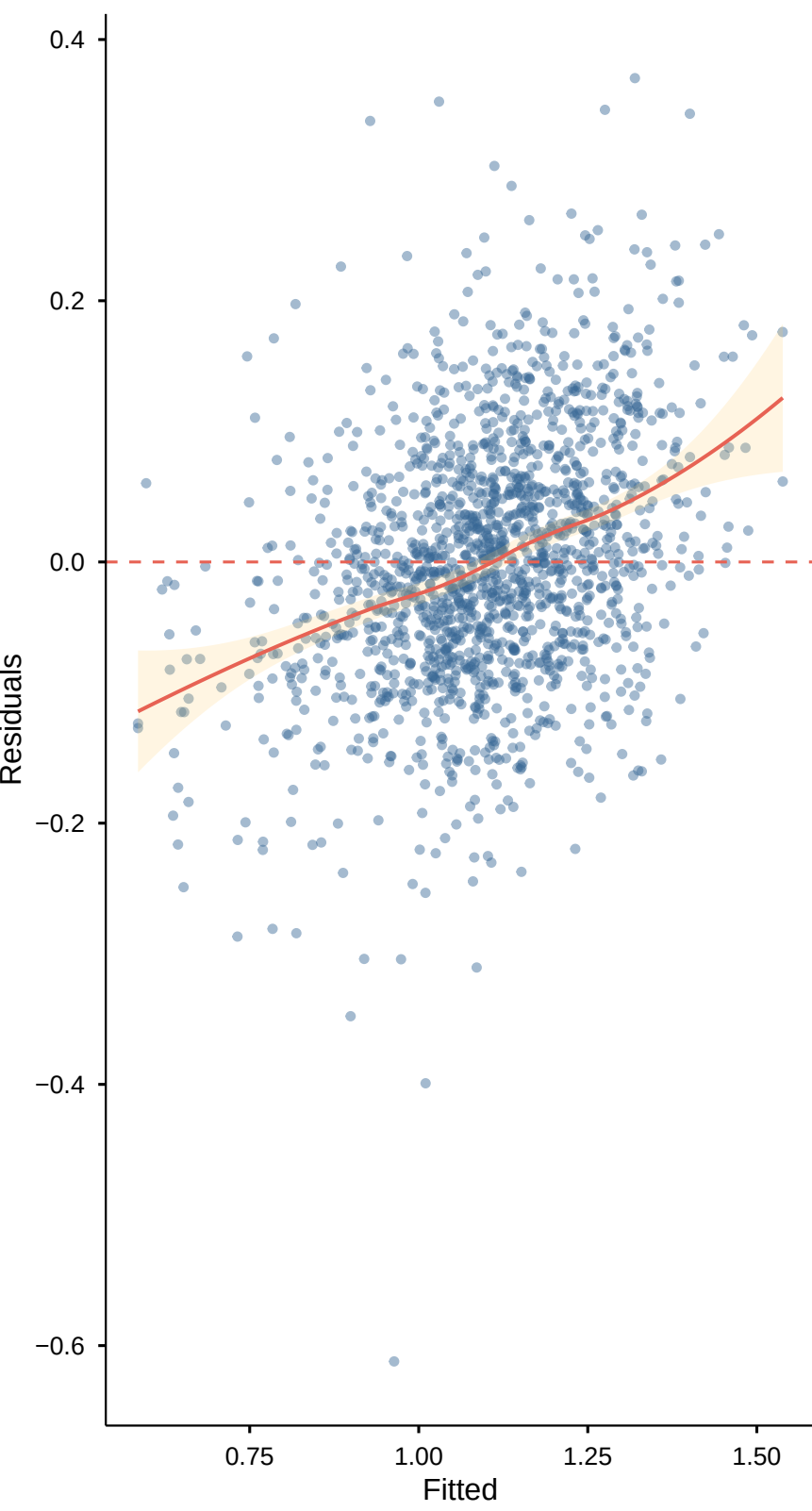


C Q-Q

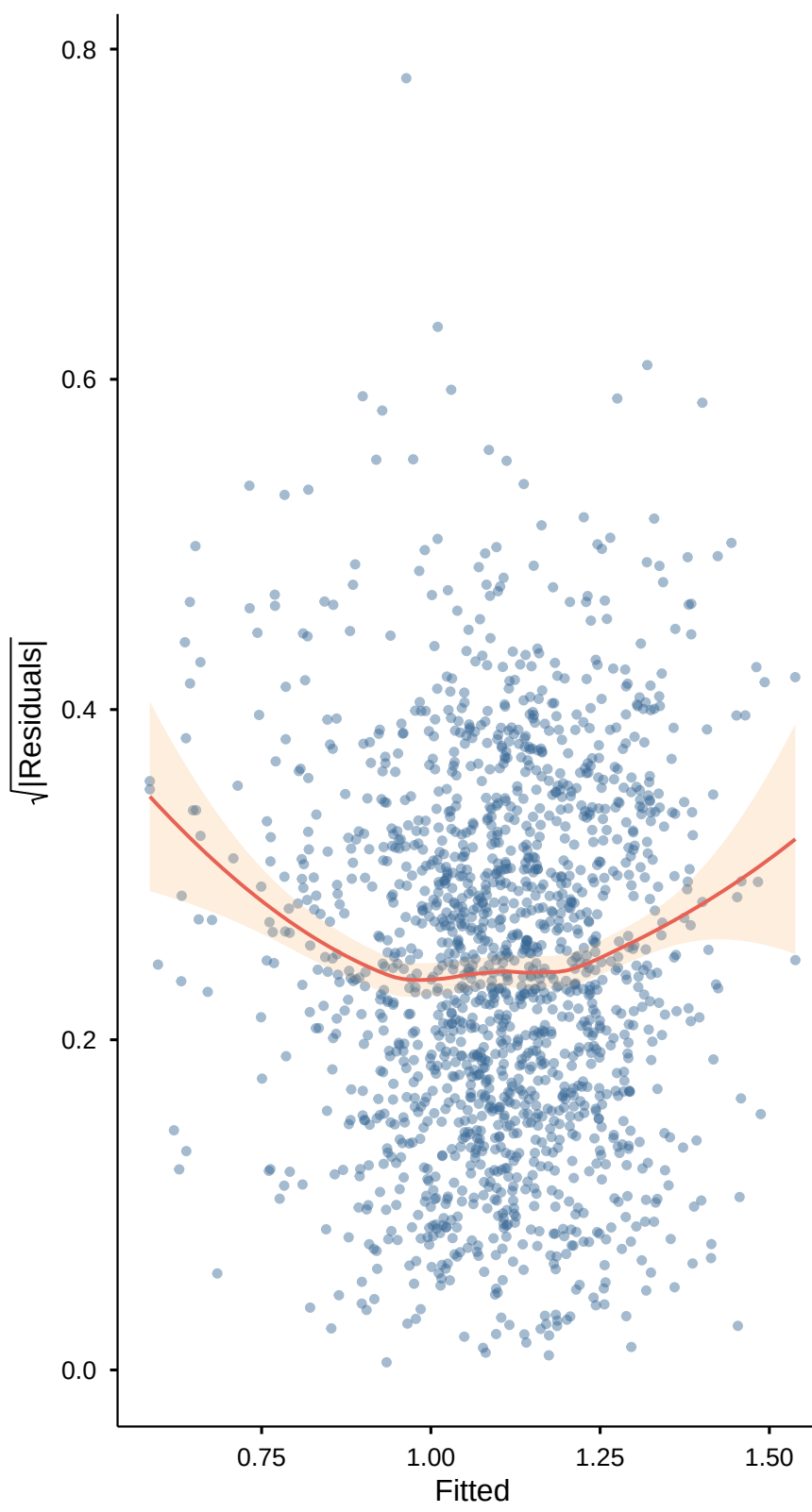


Model Assumption Checks – M5b Height + Weight, RI only

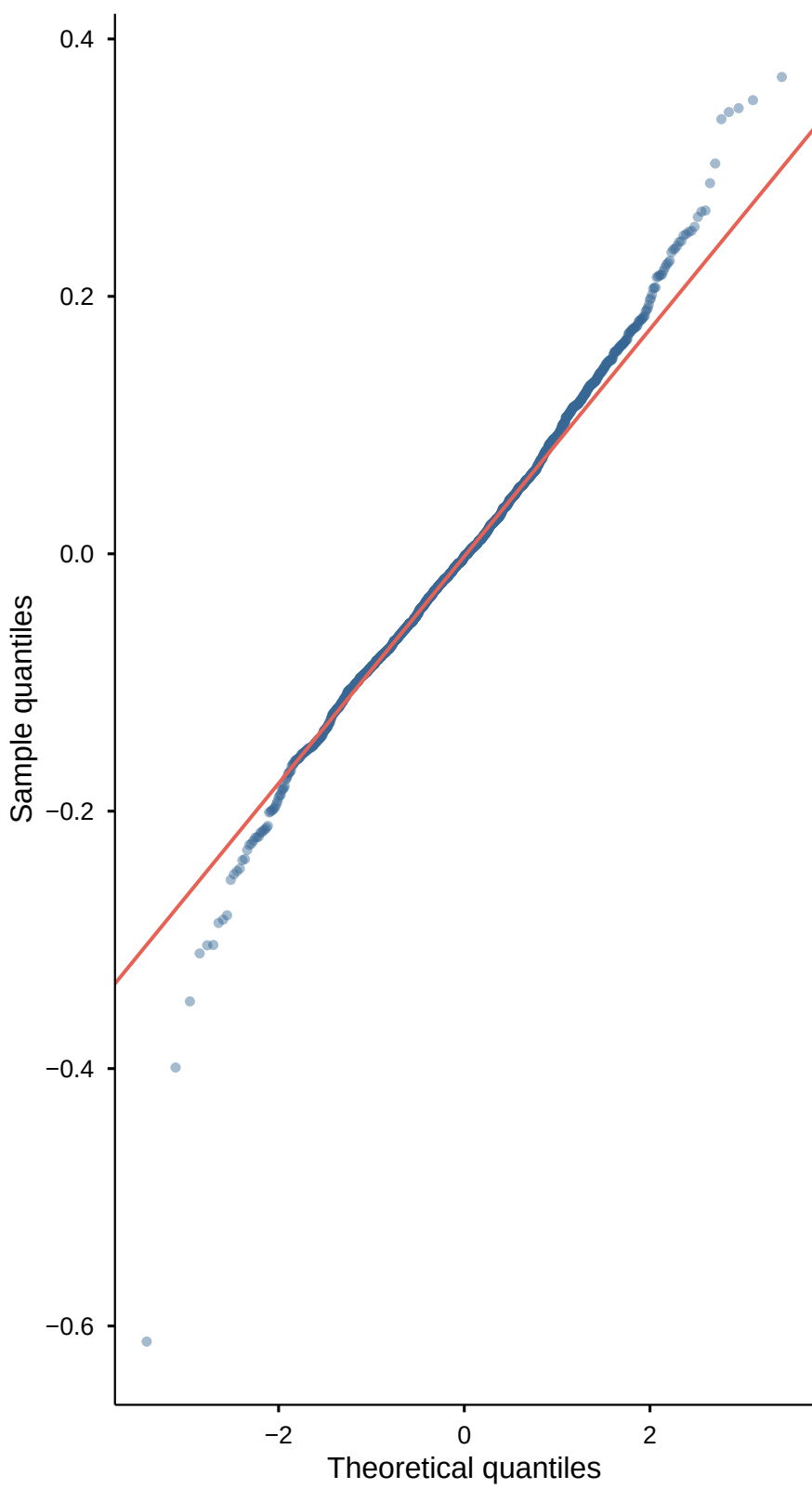
A Residuals vs Fitted



B Scale-Location

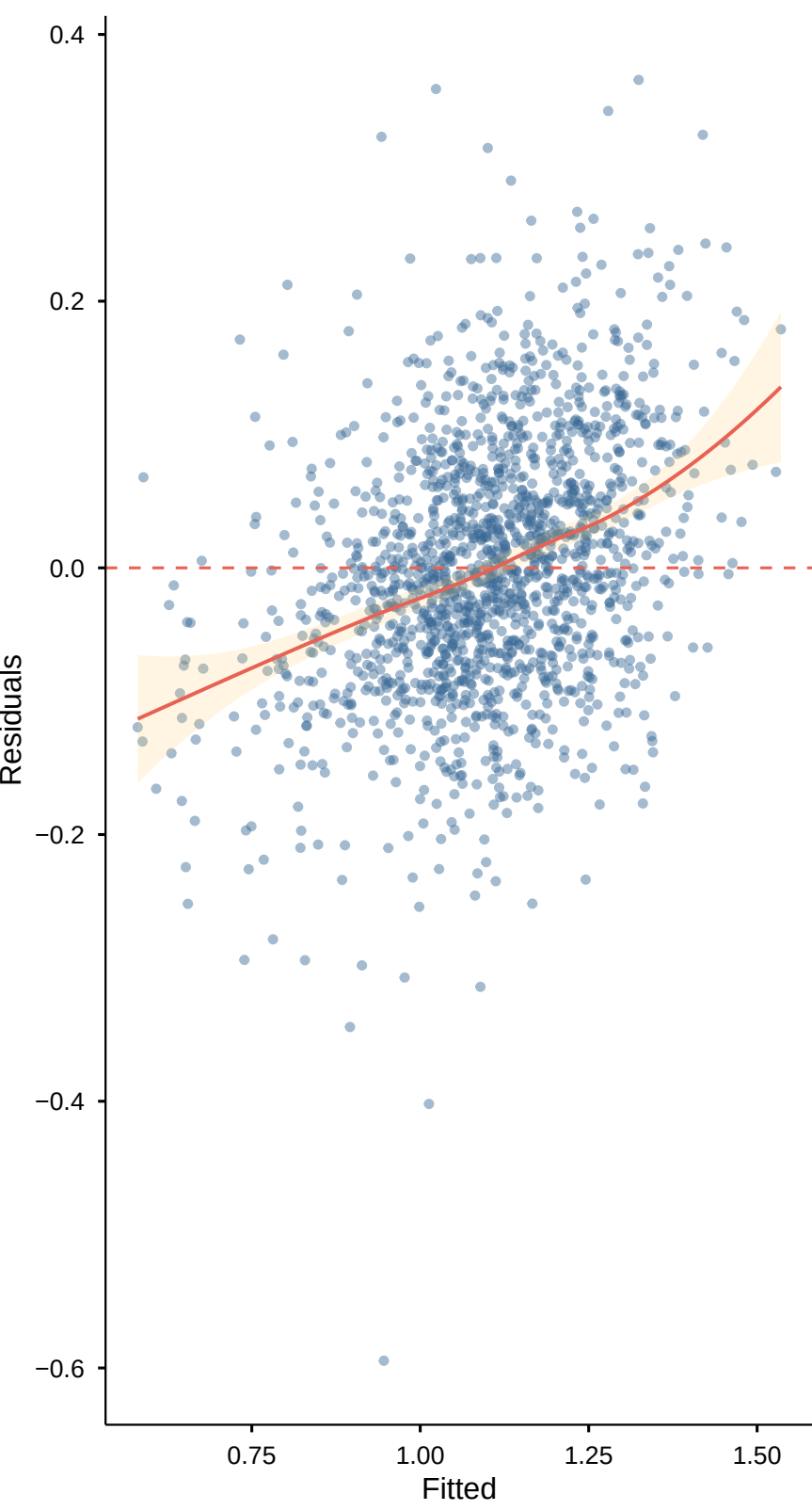


C Q-Q

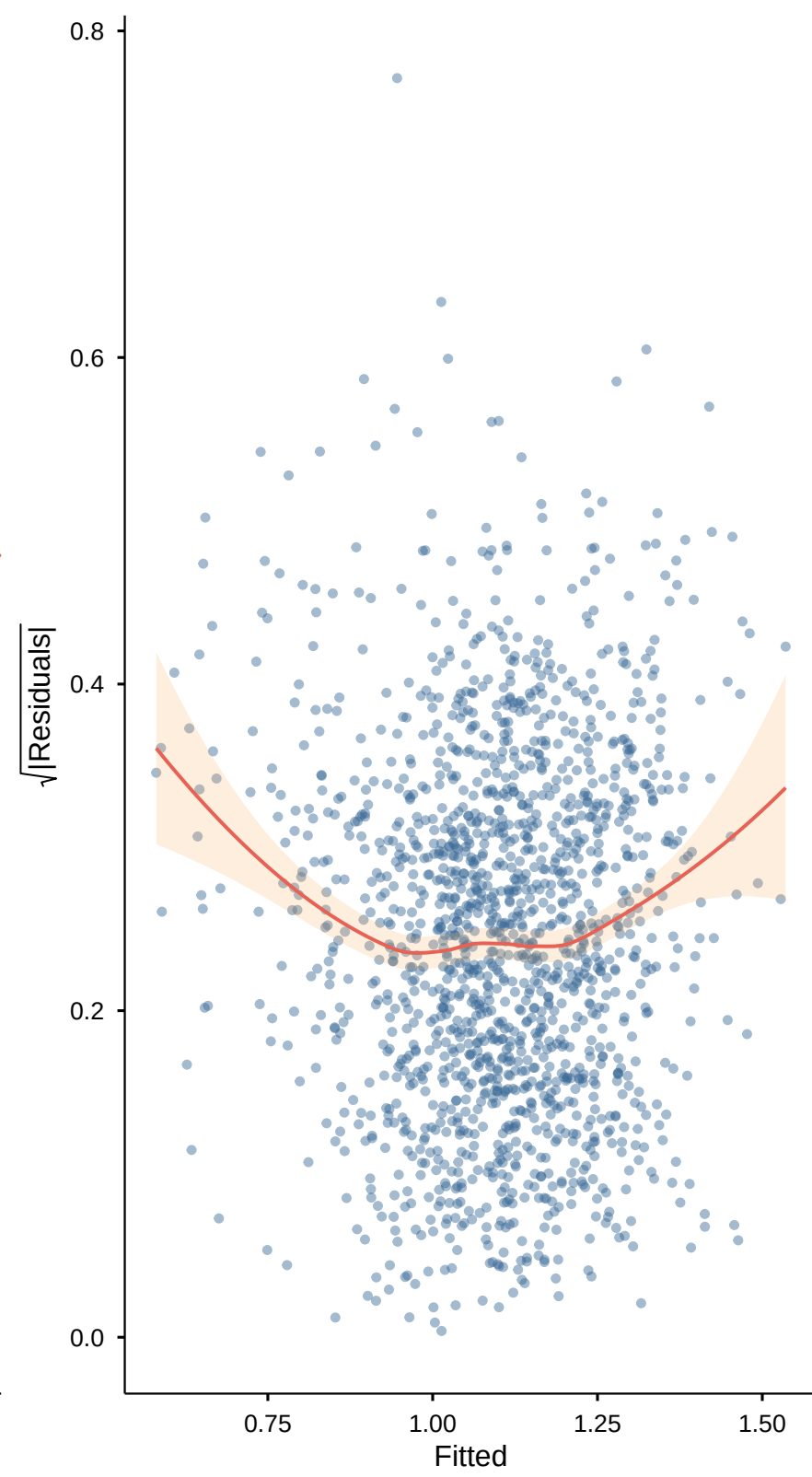


Model Assumption Checks – M6 Within-/between dairy

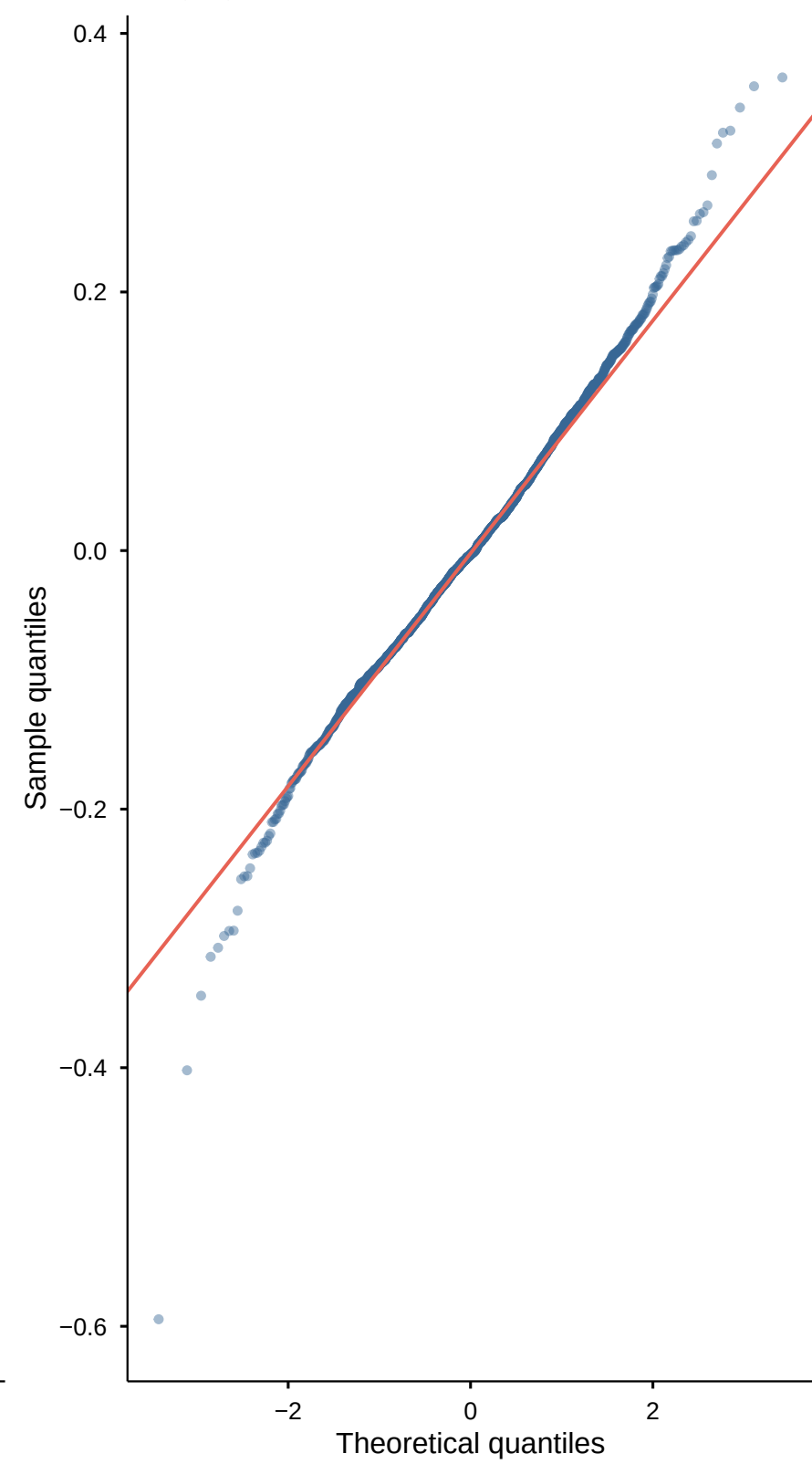
A Residuals vs Fitted



B Scale-Location

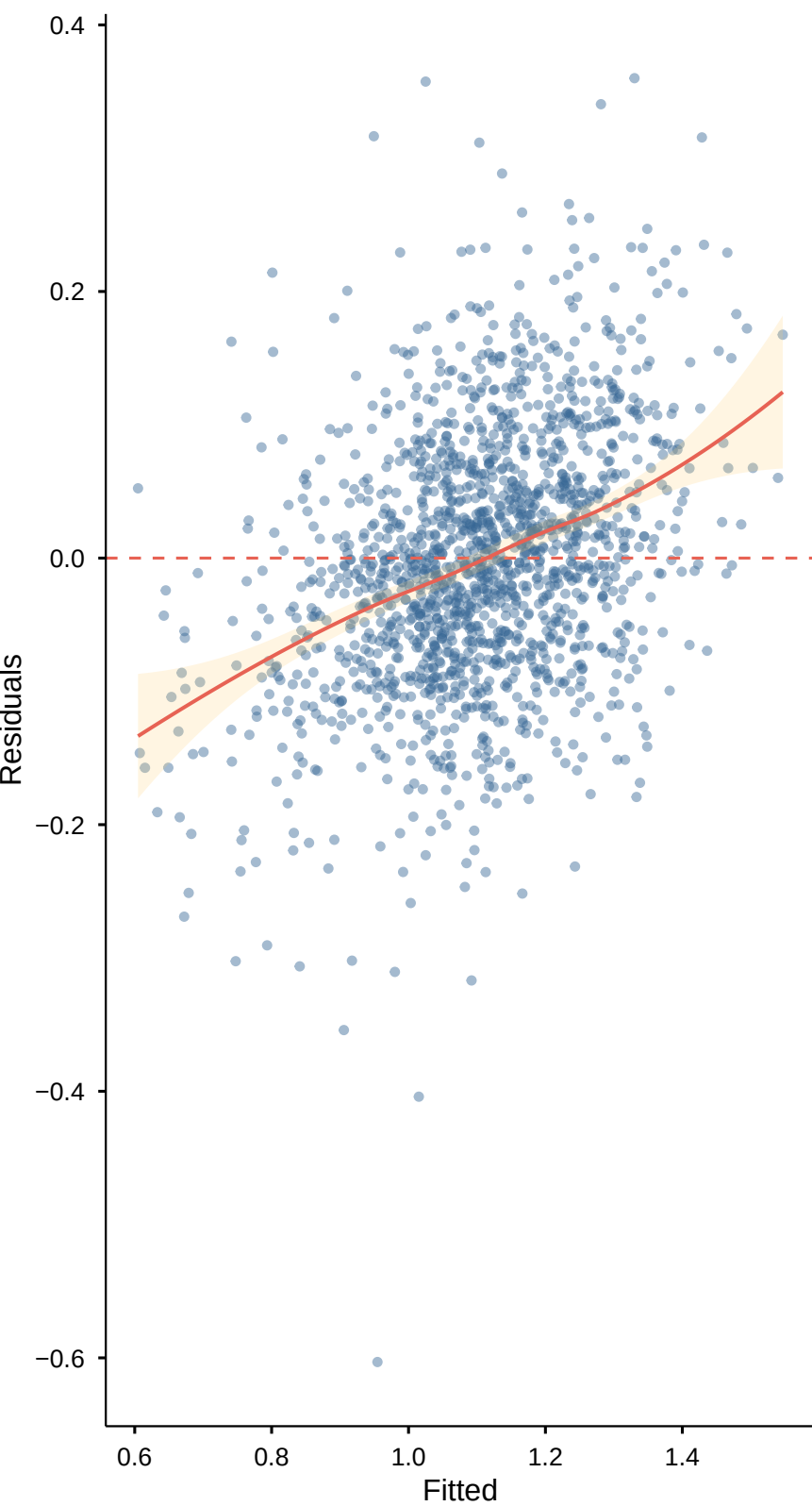


C Q-Q

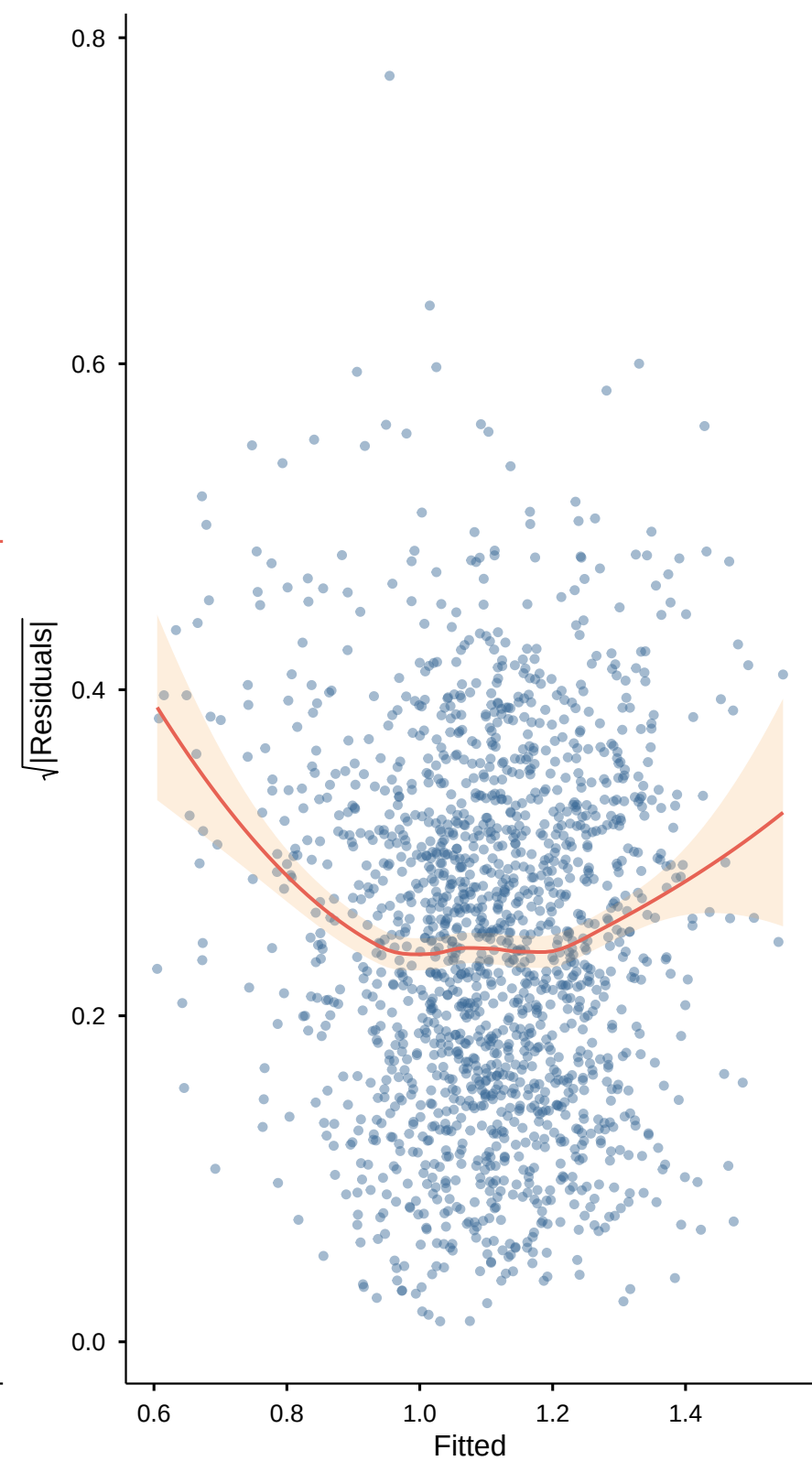


Model Assumption Checks – M8 varPower (nlme)

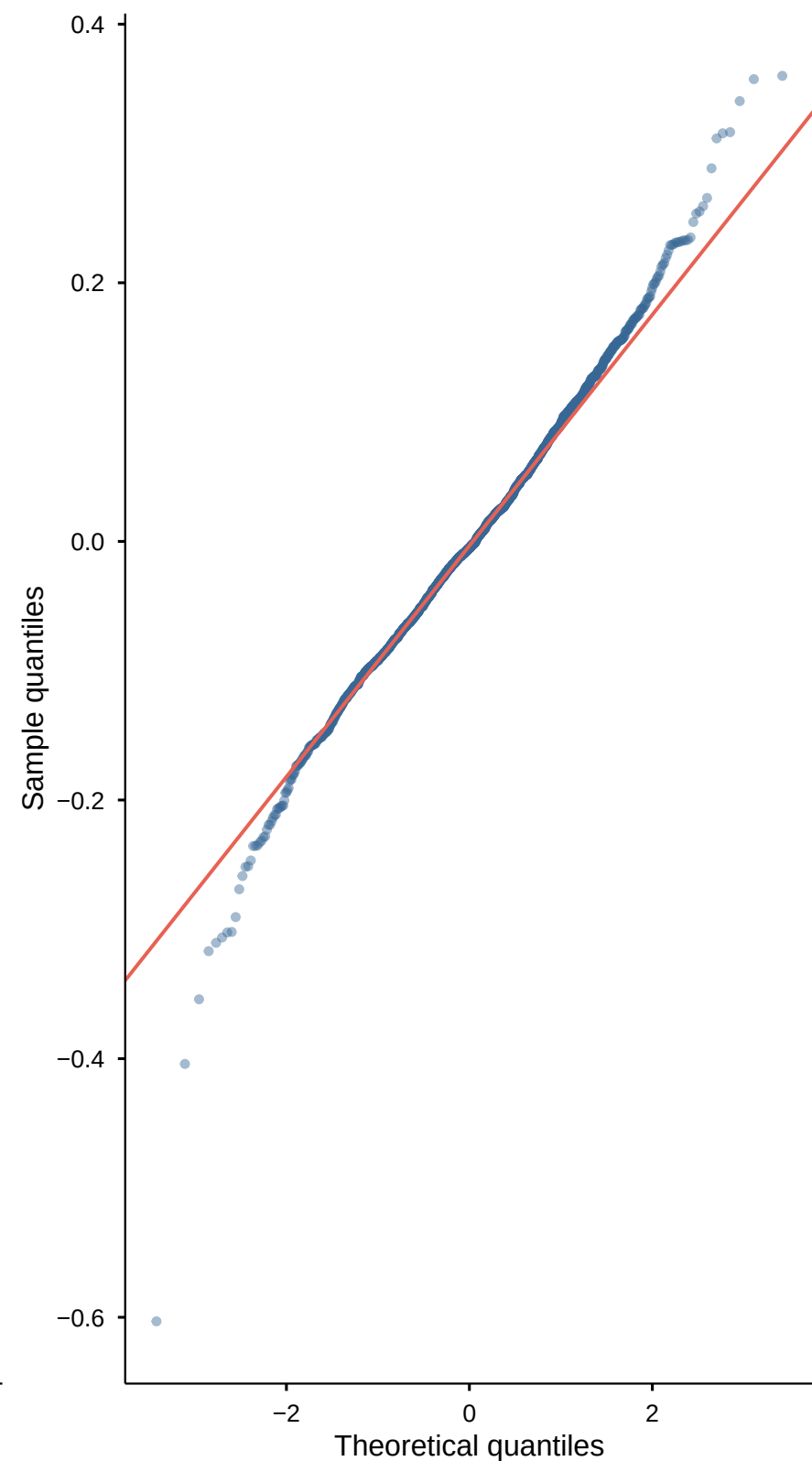
A Residuals vs Fitted



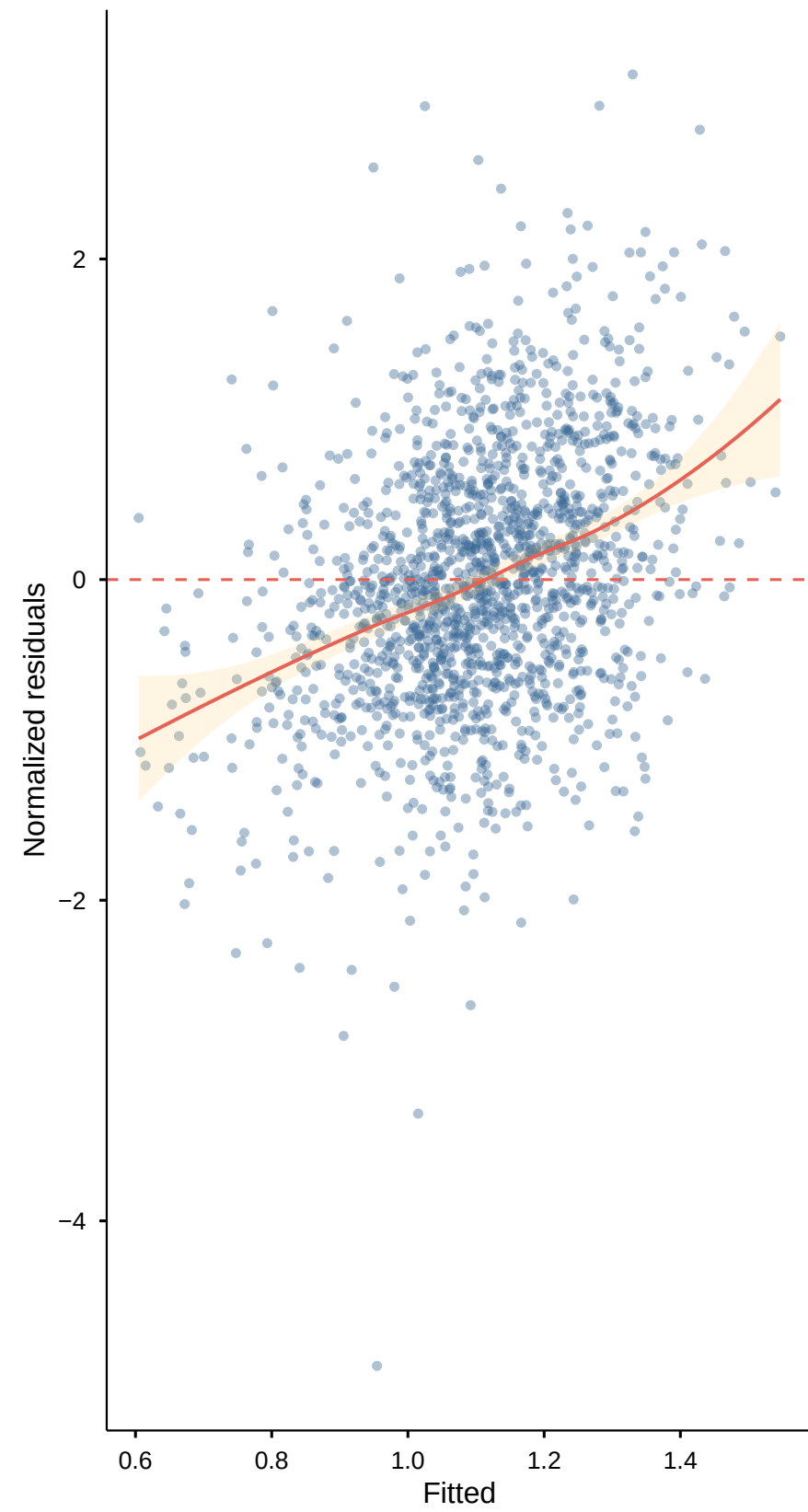
B Scale-Location



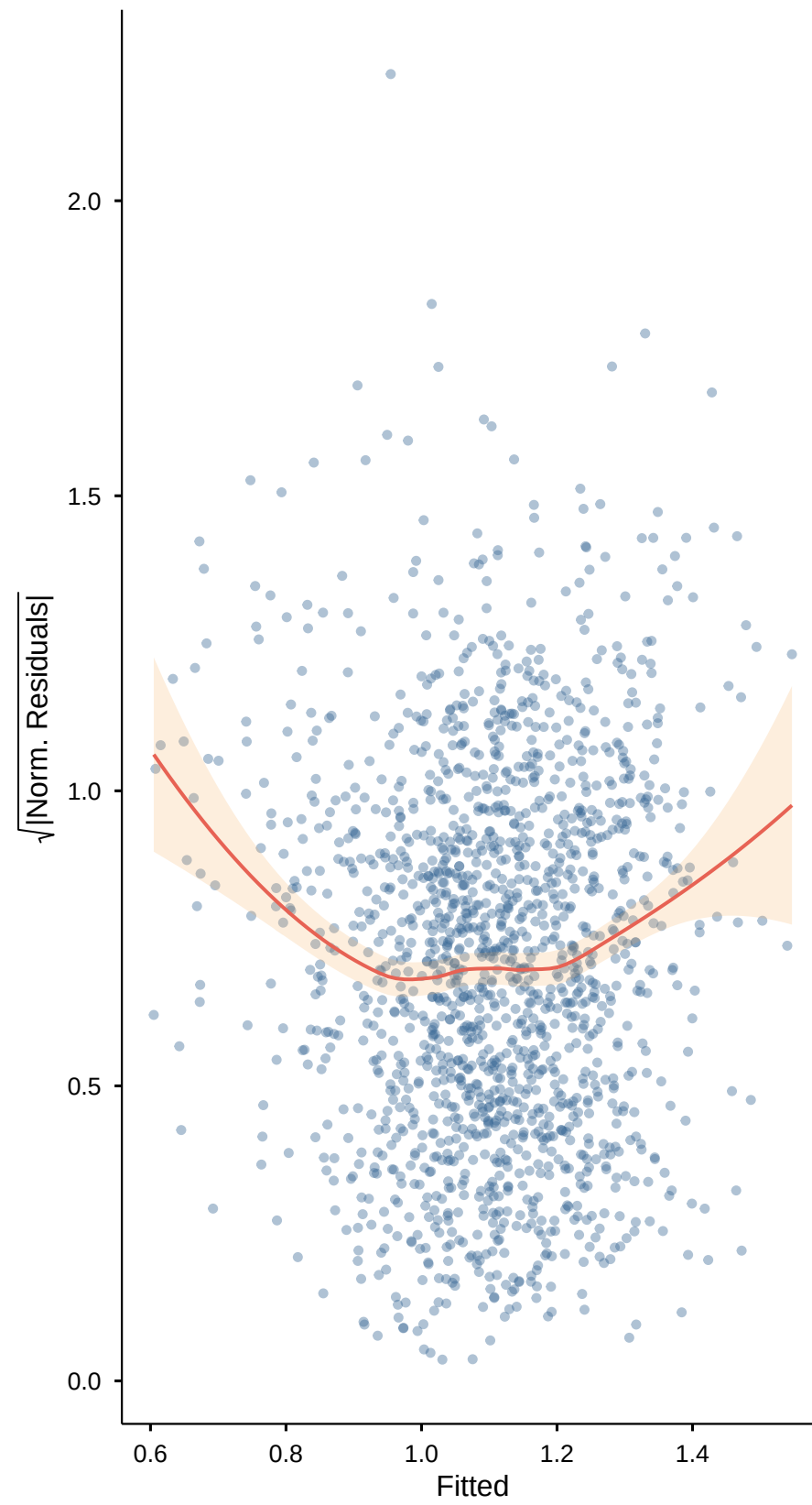
C Q-Q



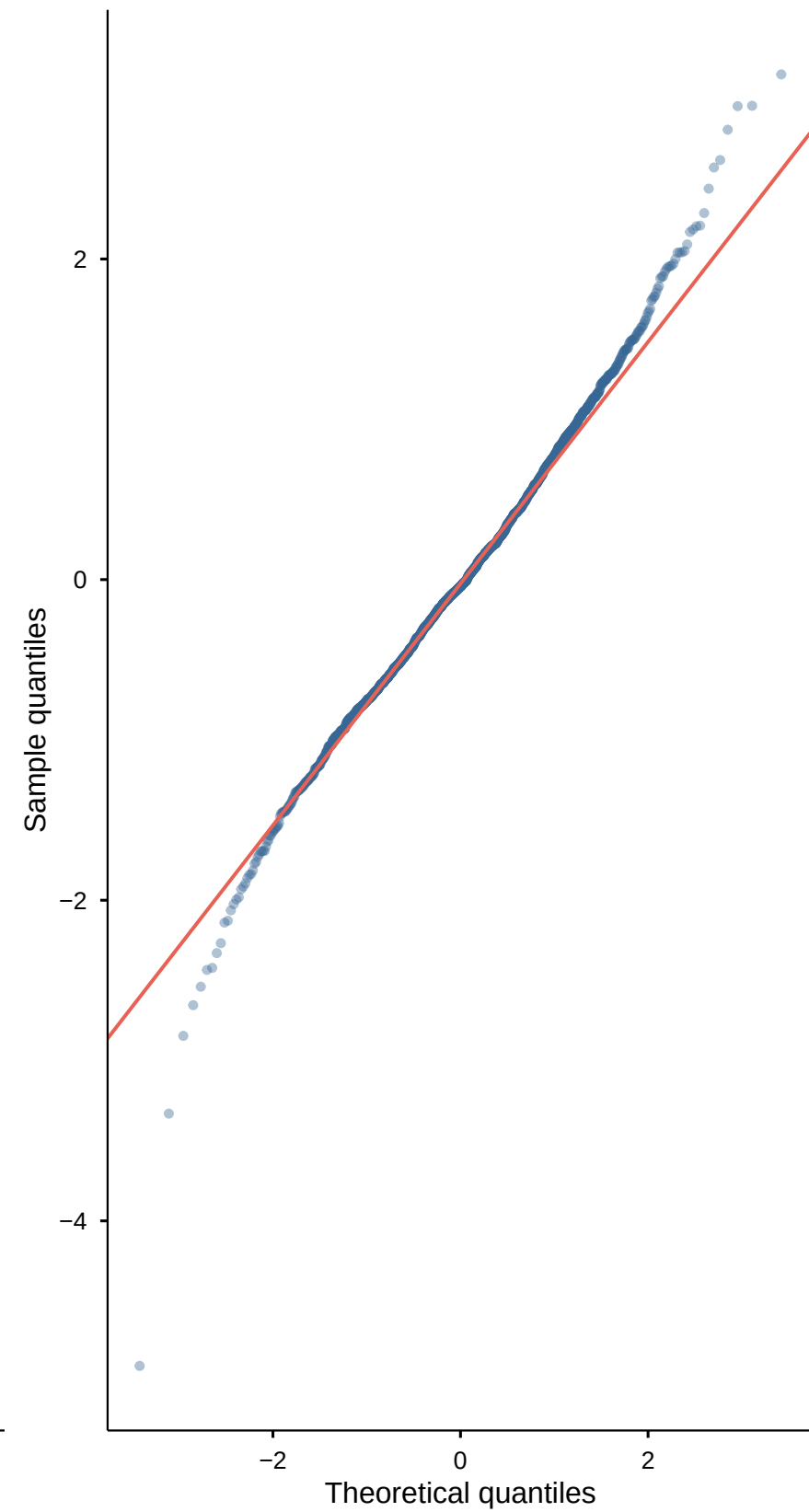
A Normalized Res. vs Fitted [varPower check]



B Scale-Location (normalized)

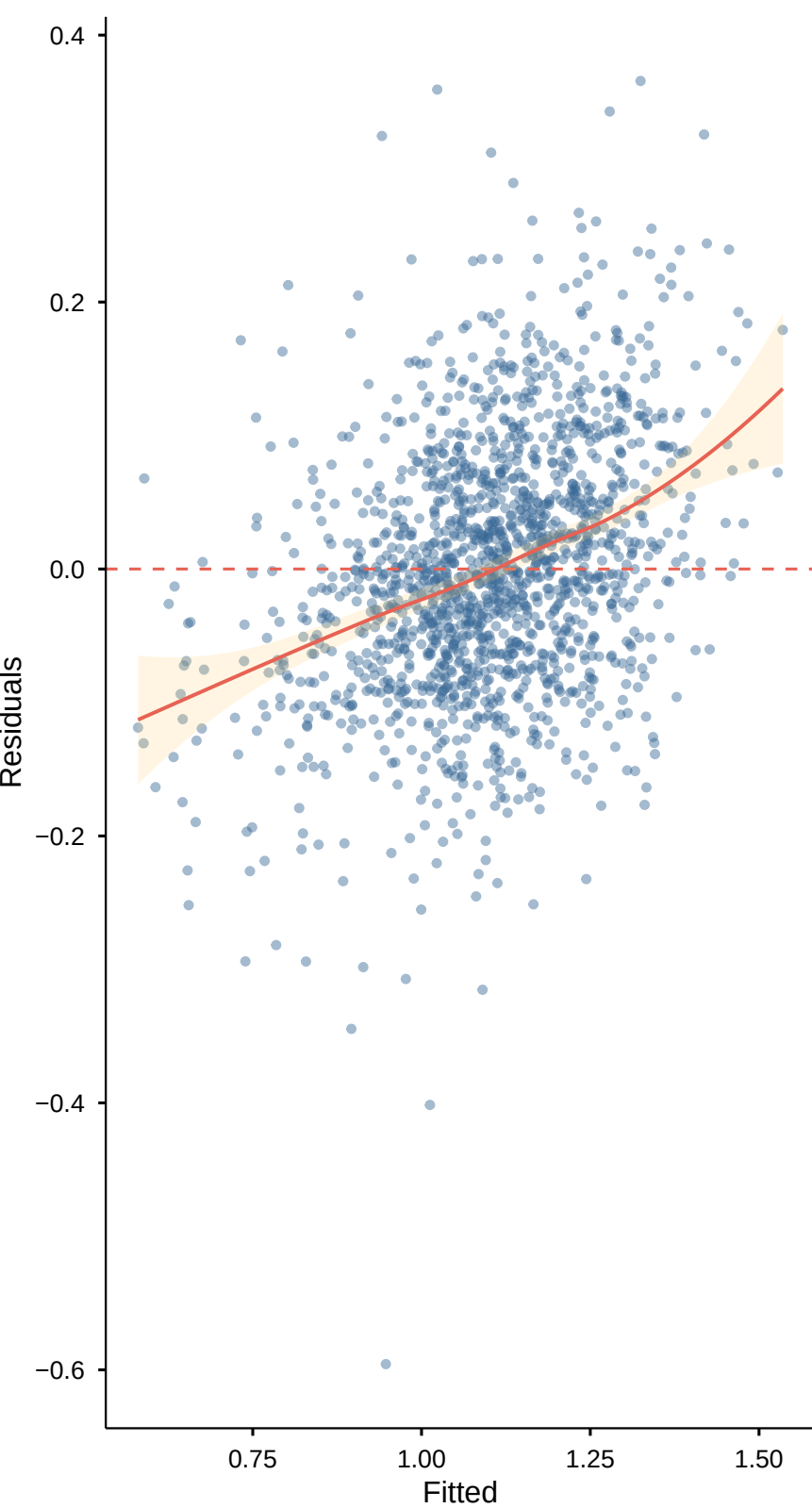


C Q-Q (normalized residuals)

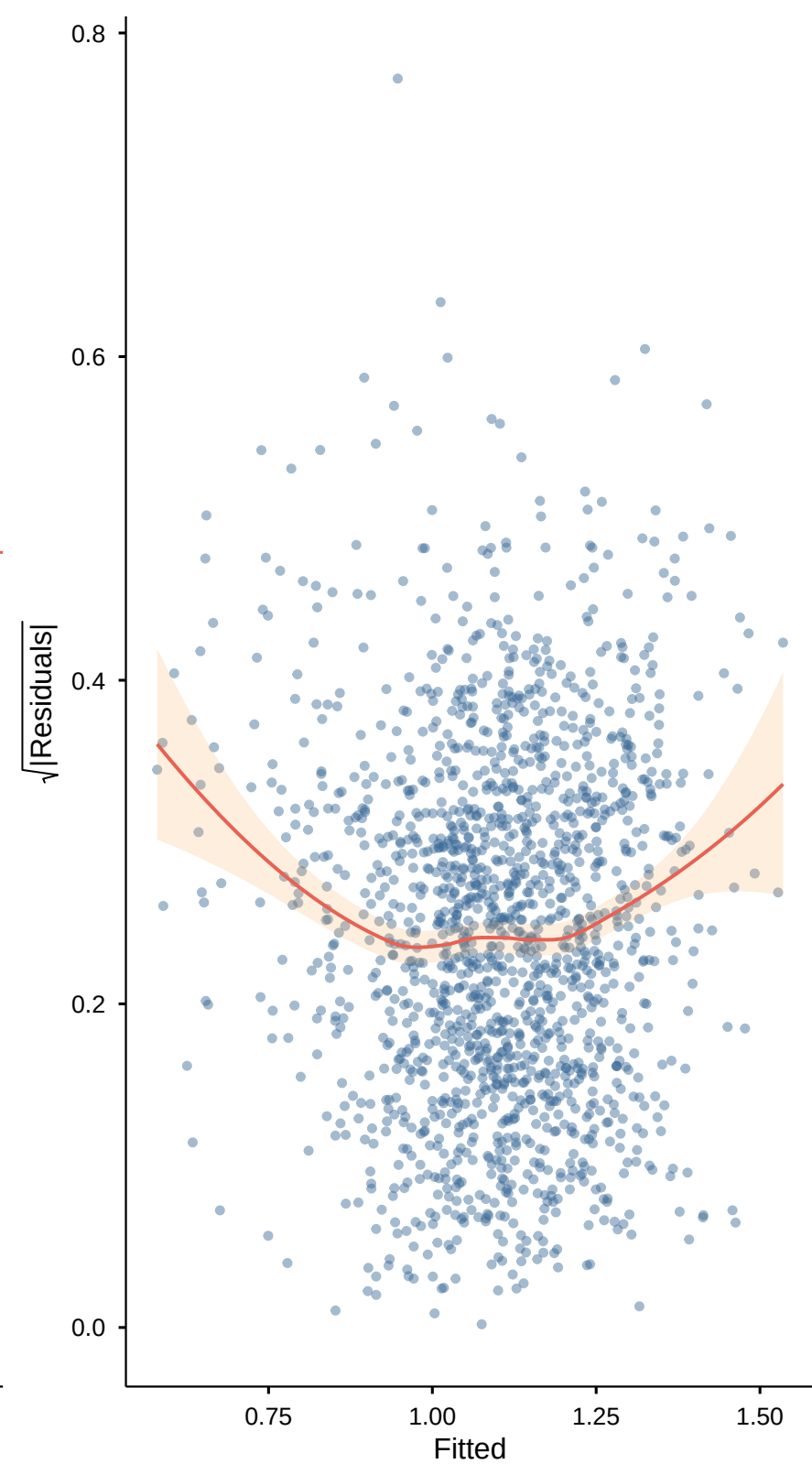


Model Assumption Checks – M9 GAMM

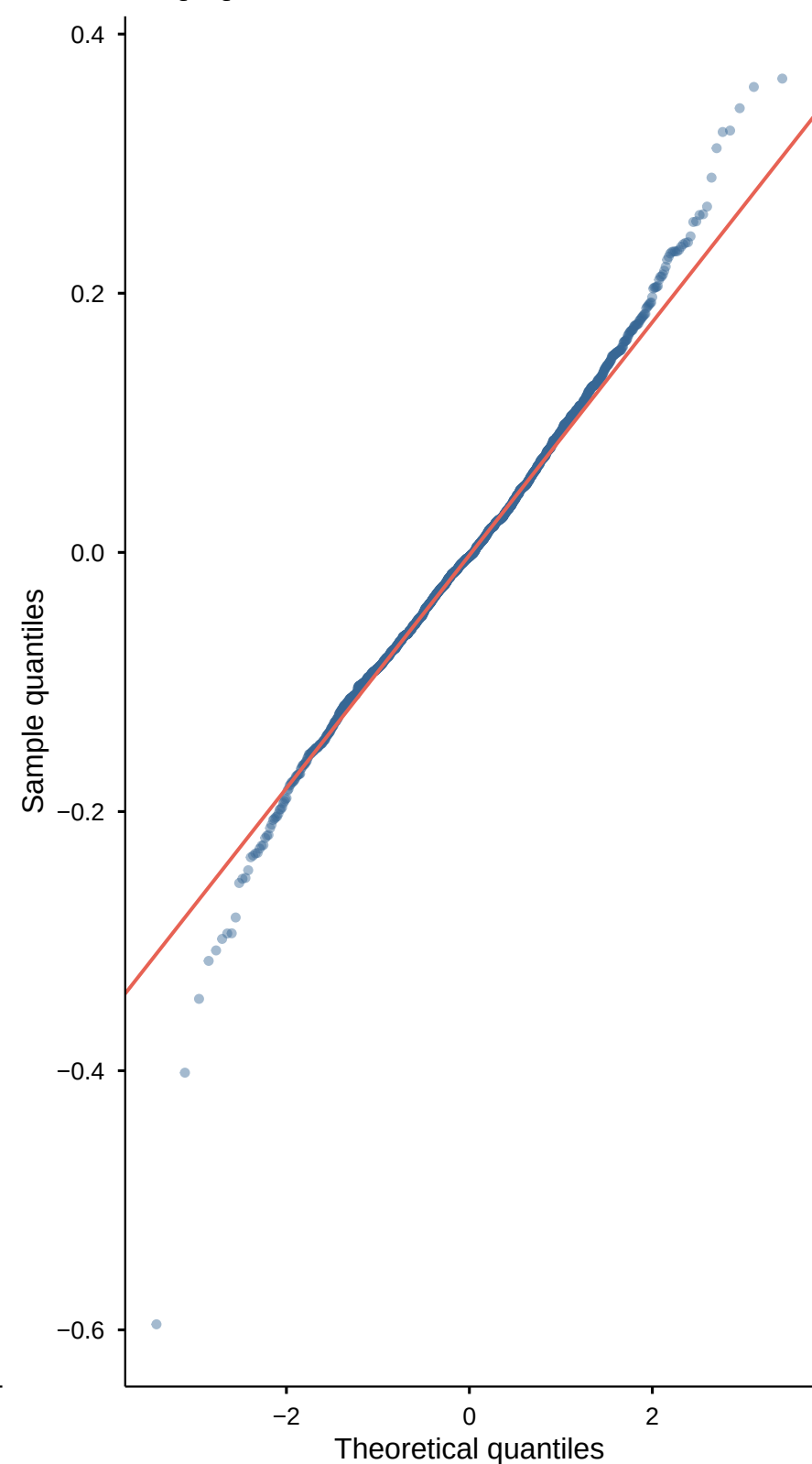
A Residuals vs Fitted



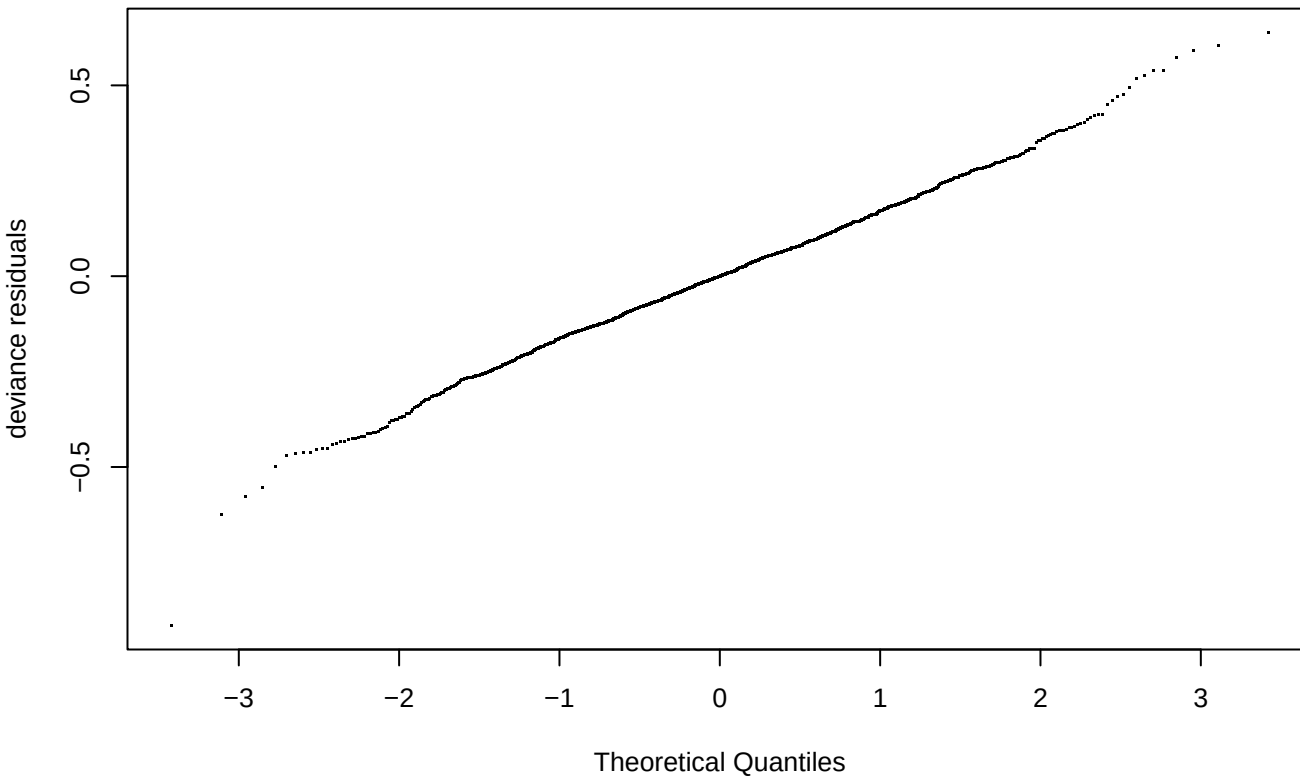
B Scale-Location



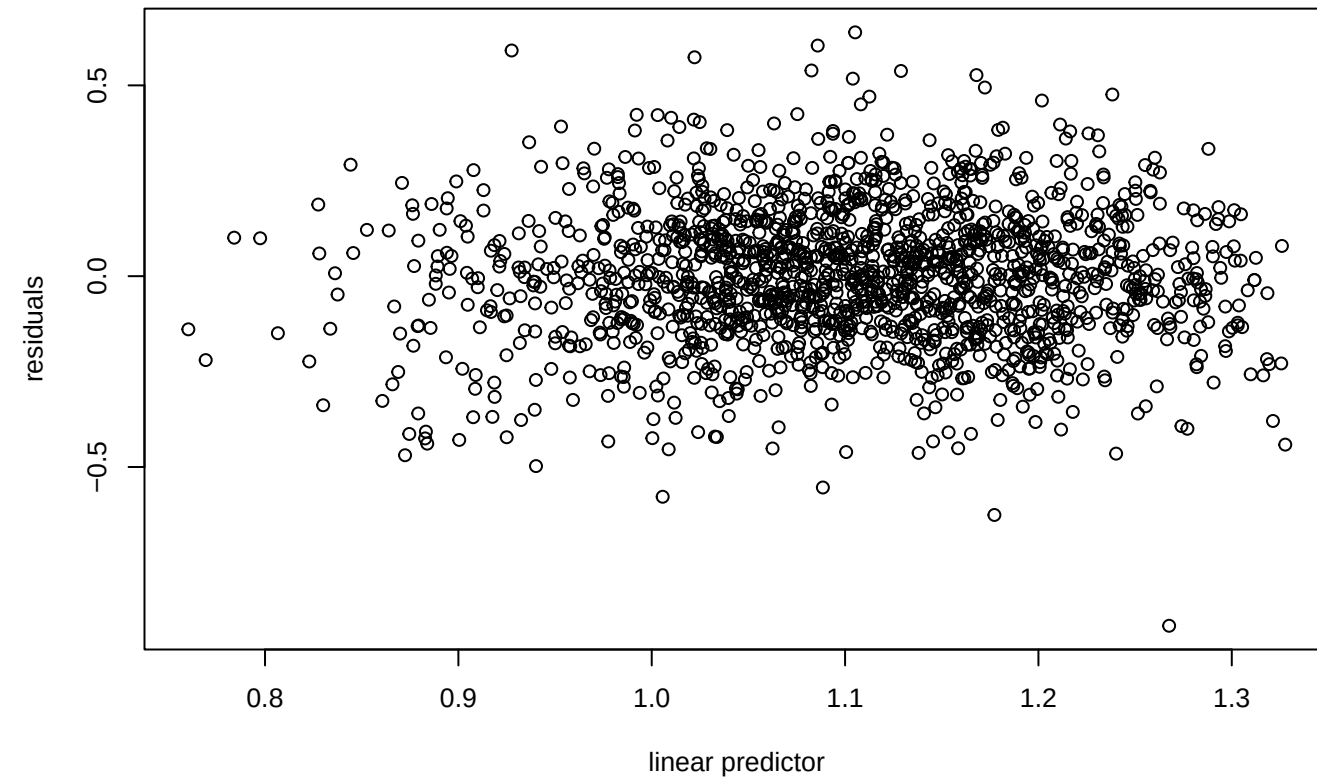
C Q-Q



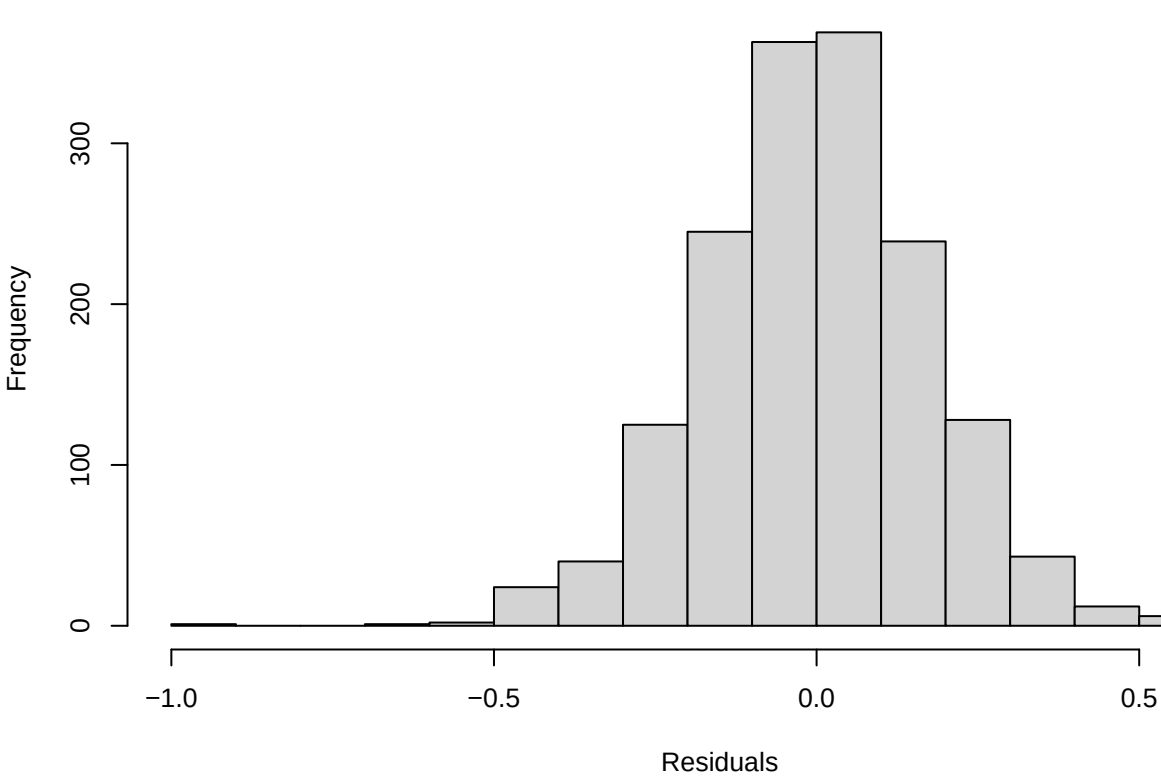
Normal Q-Q Plot



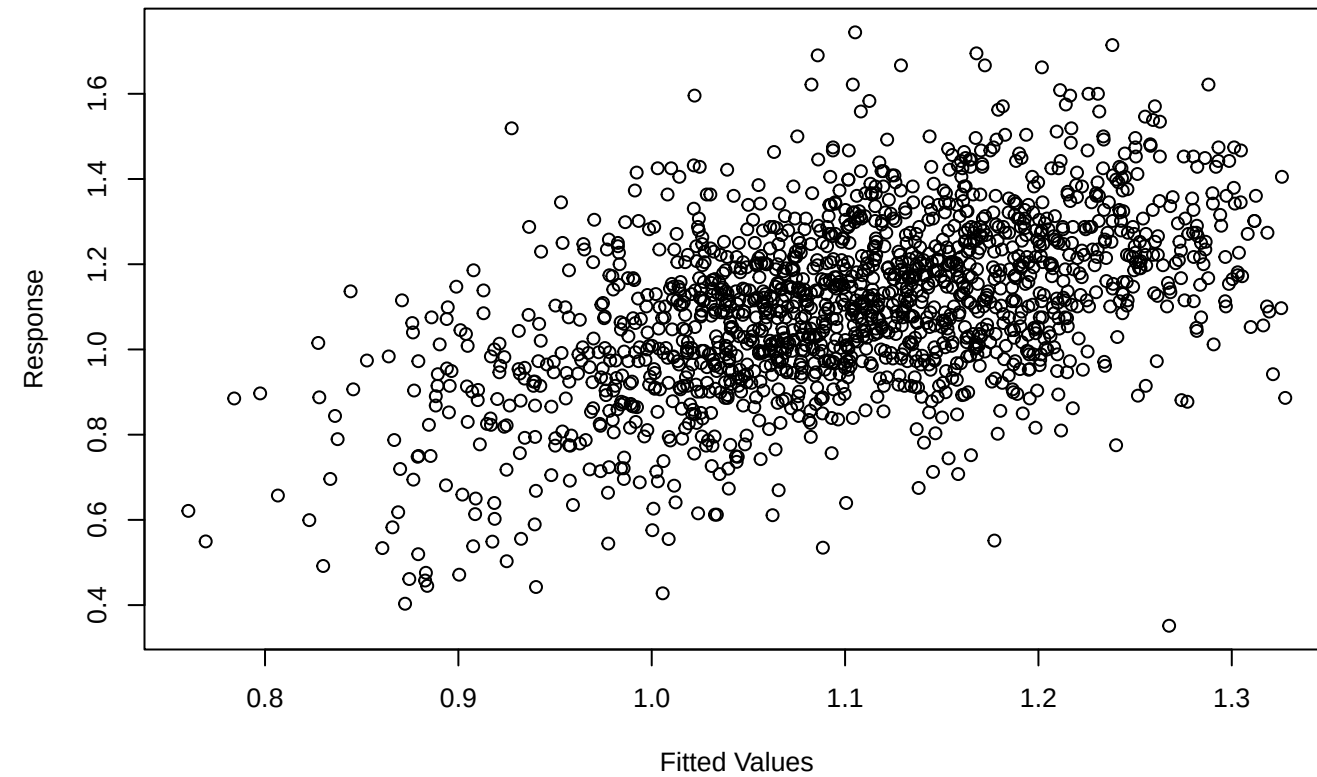
Resids vs. linear pred.



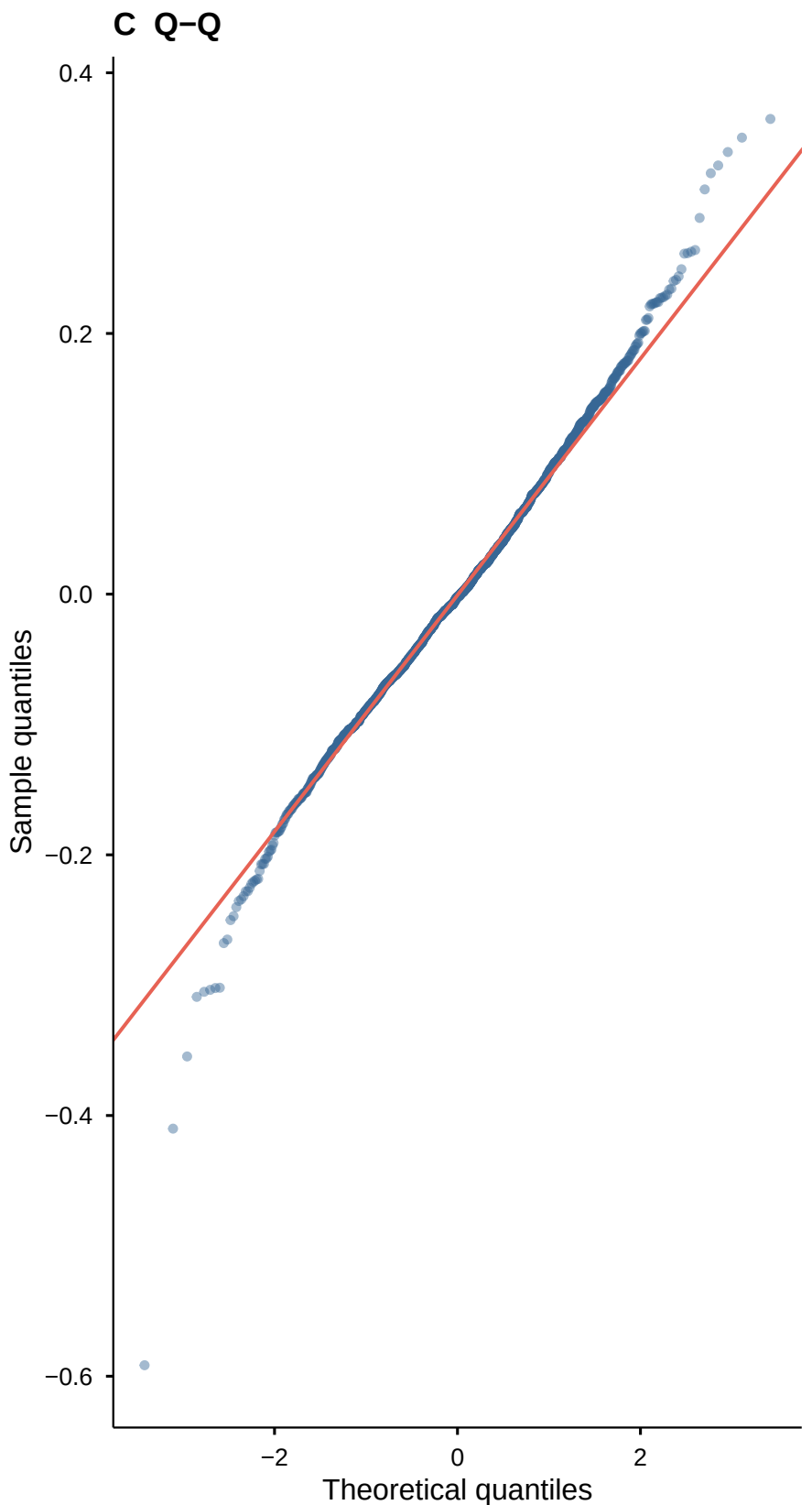
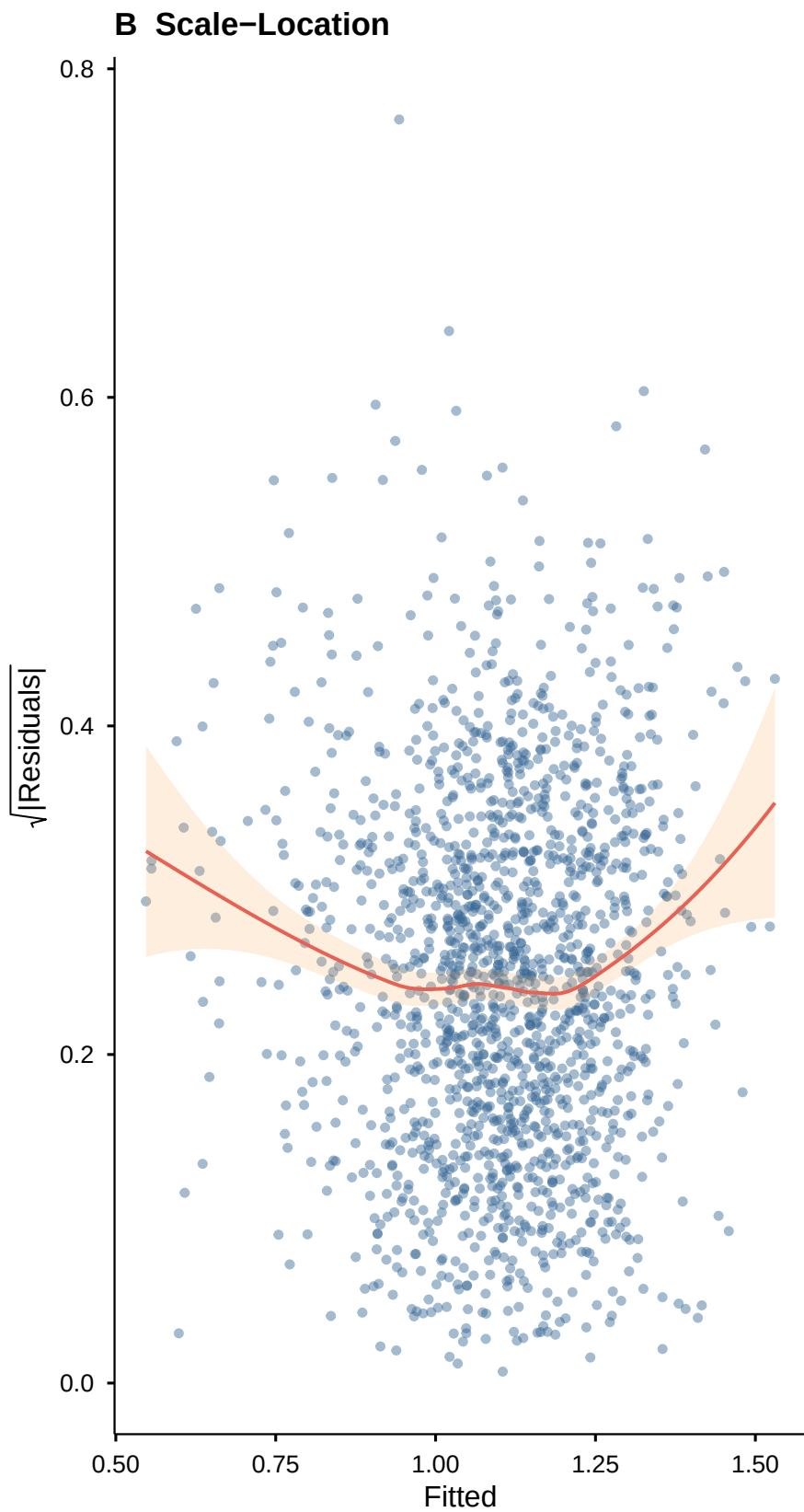
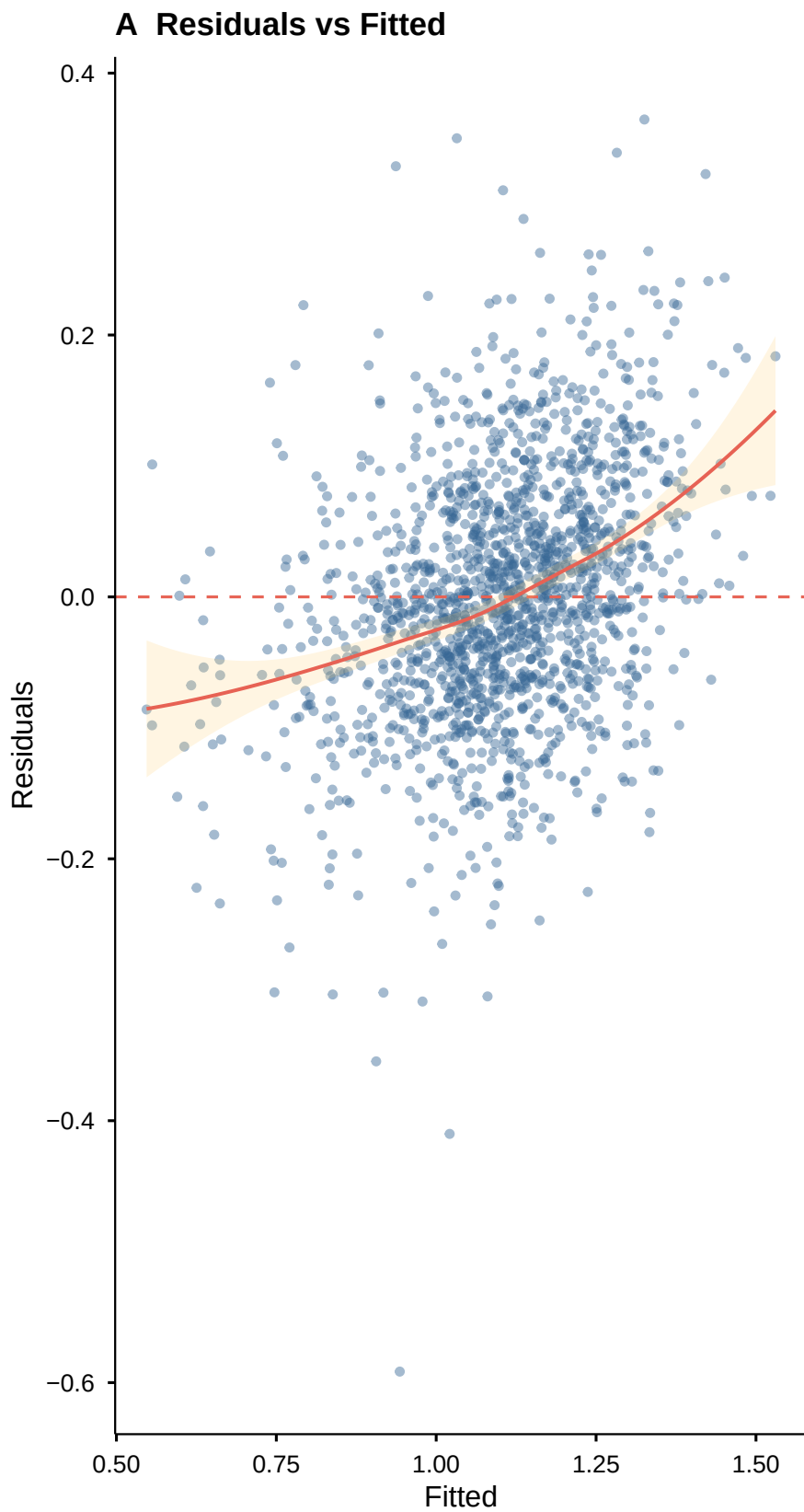
Histogram of residuals



Response vs. Fitted Values



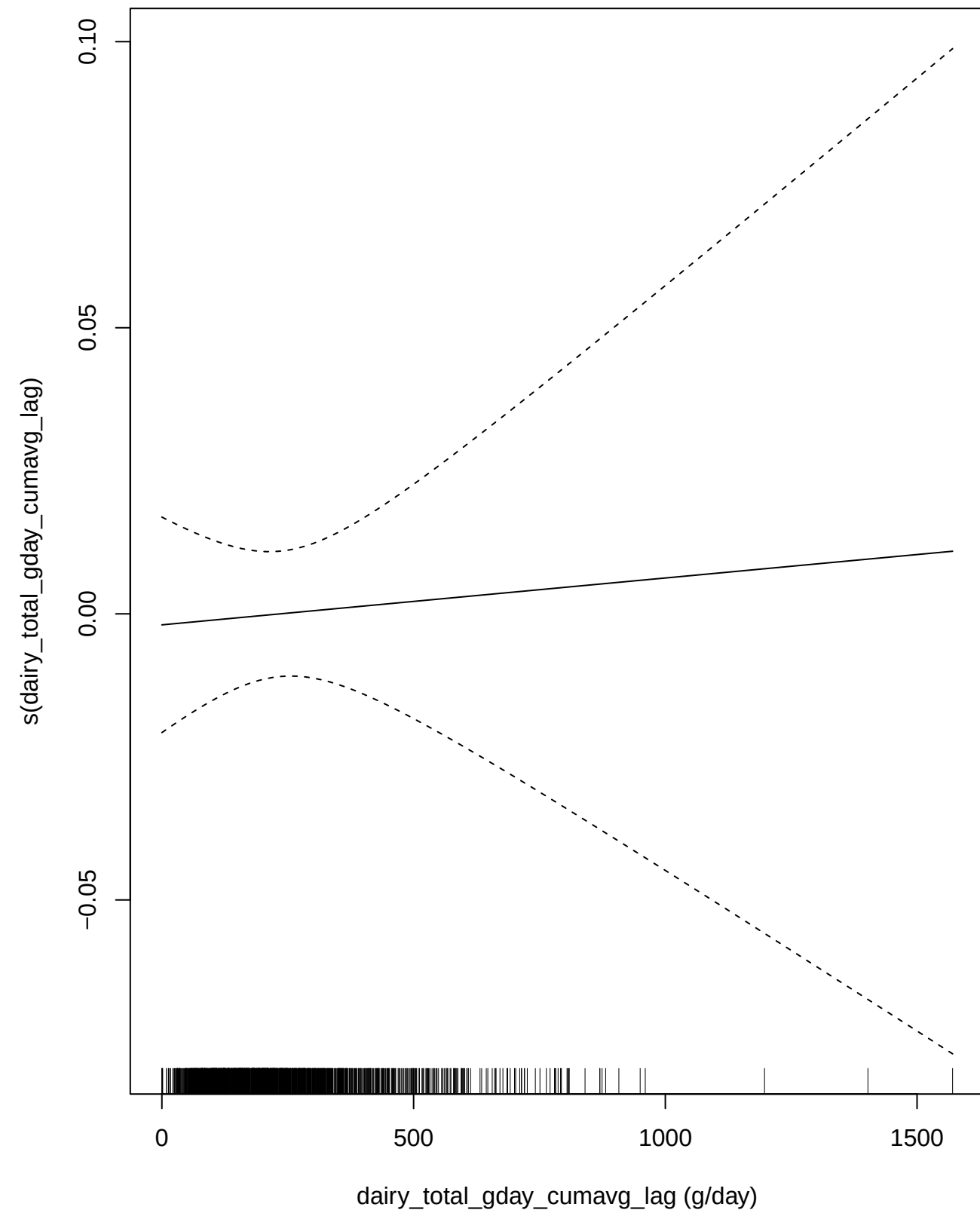
Model Assumption Checks – M10 Poly² age



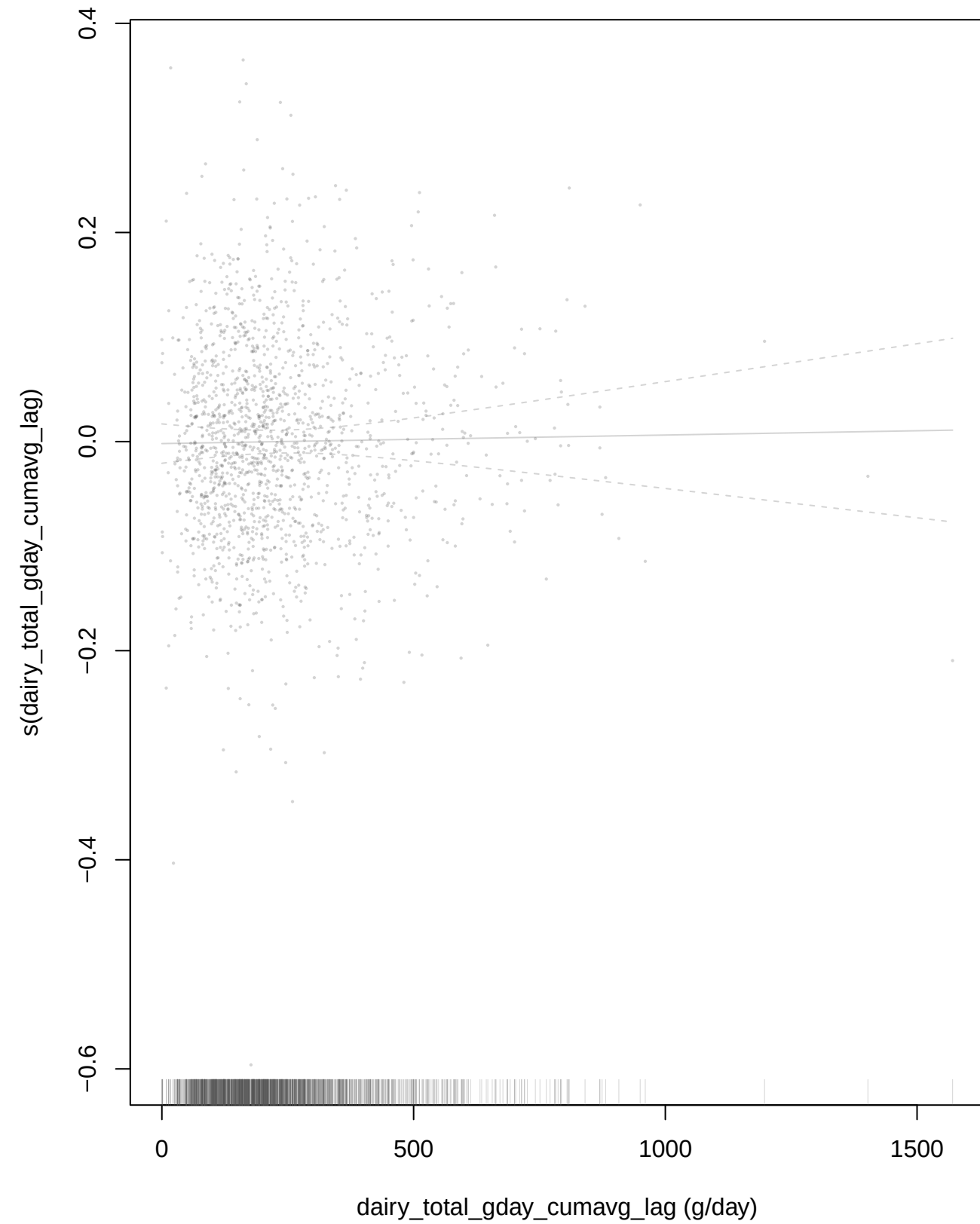
GAMM Smooth

s(dairy_total_gday_cumavg_lag, k=5) | Gait speed

GAMM smooth | gait_speed



With partial residuals



GAMM summary – Gait speed

Family: gaussian
Link function: identity

Formula:
gait_speed ~ s(dairy_total_gday_cumavg_lag, k = 5, bs = "cr") +
time_since_baseline + age_at_baseline_scaled_lag + BMI_category_lag +
education_level_lag + smoking_status_lag + pa_levels_tertile_f1_lag +
diabetes_status_lag

Parametric coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	1.147688	0.028539	40.214	< 2e-16	***
time_since_baseline	-0.003038	0.002285	-1.330	0.183781	
age_at_baseline_scaled_lag	-0.123688	0.008445	-14.646	< 2e-16	***
BMI_category_lagUnderweight	-0.013250	0.031724	-0.418	0.676252	
BMI_category_lagOverweight	-0.061723	0.010918	-5.653	1.86e-08	***
BMI_category_lagObese	-0.121083	0.015519	-7.802	1.10e-14	***
education_level_lagMedium (ISCED 3-4)	0.028034	0.016554	1.694	0.090556	.
education_level_lagHigh (ISCED 5-8)	0.055643	0.020851	2.669	0.007695	**
smoking_status_lagFormer	0.005151	0.011648	0.442	0.658386	
smoking_status_lagCurrent	-0.034027	0.016839	-2.021	0.043474	*
pa_levels_tertile_f1_lagMedium	0.002134	0.013333	0.160	0.872884	
pa_levels_tertile_f1_lagHigh	0.004759	0.014196	0.335	0.737473	
diabetes_status_lagDiabetes	-0.073096	0.021578	-3.387	0.000723	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Approximate significance of smooth terms:

	edf	Ref.df	F	p-value
s(dairy_total_gday_cumavg_lag)	1	1	0.063	0.802

R-sq.(adj) = 0.251
Scale est. = 0.014154 n = 1600

GAMM (lme component) summary – Gait speed

Linear mixed-effects model fit by maximum likelihood

Data: strip.offset(mf)
AIC BIC logLik
-1267.677 -1176.256 650.8387

Random effects:

Formula: ~Xr - 1 | g
Structure: pdIdnot

Xr1 Xr2 Xr3
StdDev: 3.757367e-06 3.757367e-06 3.757367e-06

Formula: ~1 | pt %in% g
(Intercept) Residual
StdDev: 0.1293037 0.1189696

Fixed effects: y ~ X - 1

	Value	Std.Error	DF	t-value
X(Intercept)	1.1476881	0.02854829	794	40.20164
Xtime_since_baseline	-0.0030383	0.00228552	792	-1.32936
Xage_at_baseline_scaled_lag	-0.1236881	0.00844777	794	-14.64151
XBMI_category_lagUnderweight	-0.0132497	0.03173365	792	-0.41753
XBMI_category_lagOverweight	-0.0617230	0.01092134	792	-5.65159
XBMI_category_lagObese	-0.1210834	0.01552413	792	-7.79969
Xeducation_level_lagMedium (ISCED 3-4)	0.0280335	0.01655877	794	1.69297
Xeducation_level_lagHigh (ISCED 5-8)	0.0556429	0.02085758	794	2.66776
Xsmoking_status_lagFormer	0.0051511	0.01165186	792	0.44209
Xsmoking_status_lagCurrent	-0.0340274	0.01684436	792	-2.02011
Xpa_levels_tertile_f1_lagMedium	0.0021336	0.01333756	794	0.15997
Xpa_levels_tertile_f1_lagHigh	0.0047592	0.01419998	794	0.33515
Xdiabetes_status_lagDiabetes	-0.0730956	0.02158494	792	-3.38642
Xs(dairy_total_gday_cumavg_lag)Fx1	0.0111597	0.04445588	792	0.25103

	p-value
X(Intercept)	0.0000
Xtime_since_baseline	0.1841
Xage_at_baseline_scaled_lag	0.0000
XBMI_category_lagUnderweight	0.6764
XBMI_category_lagOverweight	0.0000
XBMI_category_lagObese	0.0000
Xeducation_level_lagMedium (ISCED 3-4)	0.0909
Xeducation_level_lagHigh (ISCED 5-8)	0.0078

Xsmoking_status_lagFormer	0.6585			
Xsmoking_status_lagCurrent	0.0437			
Xpa_levels_tertile_f1_lagMedium	0.8729			
Xpa_levels_tertile_f1_lagHigh	0.7376			
Xdiabetes_status_lagDiabetes	0.0007			
Xs(dairy_total_gday_cumavg_lag)Fx1	0.8019			
Correlation:				
	X(Int)	Xtm_s	Xg____	XBMI__U
Xtime_since_baseline	-0.715			
Xage_at_baseline_scaled_lag	-0.036	-0.020		
XBMI_category_lagUnderweight	-0.038	-0.013	-0.028	
XBMI_category_lagOverweight	-0.212	0.034	-0.005	0.114
XBMI_category_lagObese	-0.182	-0.007	0.011	0.089
Xeducation_level_lagMedium (ISCED 3-4)	-0.514	0.002	0.002	-0.035
Xeducation_level_lagHigh (ISCED 5-8)	-0.446	0.001	0.083	-0.032
Xsmoking_status_lagFormer	-0.191	-0.017	0.078	0.014
Xsmoking_status_lagCurrent	-0.213	0.041	0.156	0.024
Xpa_levels_tertile_f1_lagMedium	-0.282	0.002	-0.017	0.071
Xpa_levels_tertile_f1_lagHigh	-0.295	0.001	0.069	-0.002
Xdiabetes_status_lagDiabetes	0.004	-0.046	-0.050	-0.005
Xs(dairy_total_gday_cumavg_lag)Fx1	0.020	0.021	-0.029	0.003
	XBMI_ctgry_lg0v	XBMI_ctgry_lg0b	X__M(3	
Xtime_since_baseline				
Xage_at_baseline_scaled_lag				
XBMI_category_lagUnderweight				
XBMI_category_lagOverweight				
XBMI_category_lagObese	0.432			
Xeducation_level_lagMedium (ISCED 3-4)	0.022	0.062		
Xeducation_level_lagHigh (ISCED 5-8)	0.078	0.121	0.681	
Xsmoking_status_lagFormer	-0.064	-0.037	0.016	
Xsmoking_status_lagCurrent	0.054	0.076	0.007	
Xpa_levels_tertile_f1_lagMedium	0.024	0.053	0.013	
Xpa_levels_tertile_f1_lagHigh	0.031	0.054	0.055	
Xdiabetes_status_lagDiabetes	-0.132	-0.188	0.021	
Xs(dairy_total_gday_cumavg_lag)Fx1	-0.010	-0.001	-0.042	
	X__H(5	Xsm__F	Xsm__C	X__1_M X__1_H
Xtime_since_baseline				
Xage_at_baseline_scaled_lag				
XBMI_category_lagUnderweight				
XBMI_category_lagOverweight				
XBMI_category_lagObese				
Xeducation_level_lagMedium (ISCED 3-4)				
Xeducation_level_lagHigh (ISCED 5-8)				

Xsmoking_status_lagFormer	0.002				
Xsmoking_status_lagCurrent	0.018	0.418			
Xpa_levels_tertile_f1_lagMedium	0.027	-0.015	0.059		
Xpa_levels_tertile_f1_lagHigh	0.134	-0.018	0.035	0.524	
Xdiabetes_status_lagDiabetes	0.047	-0.018	-0.037	0.044	0.028
Xs(dairy_total_gday_cumavg_lag)Fx1	-0.008	-0.018	0.019	-0.029	-0.052
Xdb__D					

Xtime_since_baseline

Xage_at_baseline_scaled_lag

XBMI_category_lagUnderweight

XBMI_category_lagOverweight

XBMI_category_lagObese

Xeducation_level_lagMedium (ISCED 3-4)

Xeducation_level_lagHigh (ISCED 5-8)

Xsmoking_status_lagFormer

Xsmoking_status_lagCurrent

Xpa_levels_tertile_f1_lagMedium

Xpa_levels_tertile_f1_lagHigh

Xdiabetes_status_lagDiabetes

Xs(dairy_total_gday_cumavg_lag)Fx1

-0.001

Standardized Within-Group Residuals:

Min	Q1	Med	Q3	Max
-5.00702254	-0.52784791	-0.02717809	0.49177233	3.07337978

Number of Observations: 1600

Number of Groups:

g	pt	%in% g
1		800

Within- / Between-Person Decomposition

Mundlak decomposition | Gait speed

Within-/Between-Person Dairy - gait_speed

Outcome: gait_speed

dairy_within_100g : deviation from subject's own mean (within-person effect)
dairy_pmean_100g : subject-level mean relative to cohort median (between-person effect)

A significant divergence between within and between estimates implies confounding by stable, unmeasured subject-level characteristics.

effect	term	estimate	std.error	statistic	df	p.value
fixed	dairy_within_100g	0.0046	0.0101	0.4557010	795.1939	0.649
fixed	dairy_pmean_100g	0.0004	0.0034	0.1107353	794.4992	0.912
conf.low	conf.high					
-0.0152	0.0244					
-0.0064	0.0071					

nlme + varPower Summary

Gait speed

nlme + varPower – Gait speed

Linear mixed-effects model fit by maximum likelihood

Data:	df
	AIC BIC logLik
	-1268.96 -1177.538 651.4799

Random effects:

Formula:	~1 pt
	(Intercept) Residual
StdDev:	0.1284465 0.1217056

Variance function:

Structure:	Power of variance covariate
Formula:	~fitted(.)
Parameter estimates:	

power
-0.2215428

Fixed effects: list(fixed_formula)

	Value	Std.Error	DF	t-value
(Intercept)	1.1466109	0.028576856	794	40.12376
dairy_100g_lag	0.0008505	0.003257377	792	0.26110
time_since_baseline	-0.0028023	0.002294513	792	-1.22130
age_at_baseline_scaled_lag	-0.1225881	0.008437696	794	-14.52863
BMI_category_lagUnderweight	-0.0140808	0.031588329	792	-0.44576
BMI_category_lagOverweight	-0.0611837	0.010891755	792	-5.61743
BMI_category_lagObese	-0.1205631	0.015571890	792	-7.74235
education_level_lagMedium (ISCED 3-4)	0.0285095	0.016536091	794	1.72408
education_level_lagHigh (ISCED 5-8)	0.0554085	0.020774607	794	2.66713
smoking_status_lagFormer	0.0052231	0.011612012	792	0.44980
smoking_status_lagCurrent	-0.0340053	0.016799107	792	-2.02424
pa_levels_tertile_f1_lagMedium	0.0022598	0.013292631	794	0.17001
pa_levels_tertile_f1_lagHigh	0.0052924	0.014146622	794	0.37411
diabetes_status_lagDiabetes	-0.0741629	0.021810123	792	-3.40039

	p-value
(Intercept)	0.0000
dairy_100g_lag	0.7941
time_since_baseline	0.2223
age_at_baseline_scaled_lag	0.0000
BMI_category_lagUnderweight	0.6559
BMI_category_lagOverweight	0.0000
BMI_category_lagObese	0.0000

```
education_level_lagMedium (ISCED 3-4)  0.0851
education_level_lagHigh (ISCED 5-8)    0.0078
smoking_status_lagFormer                0.6530
smoking_status_lagCurrent               0.0433
pa_levels_tertile_f1_lagMedium          0.8650
pa_levels_tertile_f1_lagHigh            0.7084
diabetes_status_lagDiabetes              0.0007

Correlation:

(Intr) d_100_  tm_sn_ ag____ BMI__U

dairy_100g_lag                          -0.023
time_since_baseline                     -0.718  0.023
age_at_baseline_scaled_lag              -0.033 -0.029 -0.020
BMI_category_lagUnderweight              -0.038  0.003 -0.013 -0.029
BMI_category_lagOverweight               -0.210 -0.010  0.035 -0.004  0.113
BMI_category_lagObese                    -0.180 -0.002 -0.005  0.012  0.089
education_level_lagMedium (ISCED 3-4)    -0.511 -0.043  0.002  0.004 -0.035
education_level_lagHigh (ISCED 5-8)      -0.446 -0.008  0.001  0.083 -0.032
smoking_status_lagFormer                  -0.189 -0.018 -0.017  0.078  0.015
smoking_status_lagCurrent                 -0.214  0.019  0.042  0.156  0.025
pa_levels_tertile_f1_lagMedium            -0.280 -0.028  0.002 -0.017  0.072
pa_levels_tertile_f1_lagHigh              -0.292 -0.052  0.002  0.068 -0.001
diabetes_status_lagDiabetes               0.005 -0.001 -0.048 -0.049 -0.005

BMI_ctgry_lg0v BMI_ctgry_lg0b e__M(3

dairy_100g_lag
time_since_baseline
age_at_baseline_scaled_lag
BMI_category_lagUnderweight
BMI_category_lagOverweight
BMI_category_lagObese                    0.426
education_level_lagMedium (ISCED 3-4)    0.022          0.061
education_level_lagHigh (ISCED 5-8)      0.078          0.120          0.683
smoking_status_lagFormer                  -0.064          -0.037          0.016
smoking_status_lagCurrent                 0.055          0.076          0.008
pa_levels_tertile_f1_lagMedium            0.025          0.053          0.013
pa_levels_tertile_f1_lagHigh              0.031          0.053          0.056
diabetes_status_lagDiabetes               -0.131          -0.187          0.021

e__H(5 smk__F smk__C p____1_M p____1_H

dairy_100g_lag
time_since_baseline
age_at_baseline_scaled_lag
BMI_category_lagUnderweight
BMI_category_lagOverweight
BMI_category_lagObese
```

education_level_lagMedium (ISCED 3-4)
education_level_lagHigh (ISCED 5-8)
smoking_status_lagFormer 0.002
smoking_status_lagCurrent 0.018 0.417
pa_levels_tertile_f1_lagMedium 0.027 -0.015 0.059
pa_levels_tertile_f1_lagHigh 0.135 -0.018 0.035 0.524
diabetes_status_lagDiabetes 0.047 -0.018 -0.037 0.044 0.029

Standardized Within-Group Residuals:
Min Q1 Med Q3 Max
-4.90488330 -0.53382855 -0.04229008 0.48332510 3.15121436

Number of Observations: 1600
Number of Groups: 800

Session Info

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system   aarch64, darwin20
ui       X11
language (EN)
collate  en_US.UTF-8
ctype    en_US.UTF-8
tz       Europe/Zurich
date     2026-06-28
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[41m [37mP [39m [49m .. Loaded and on-disk path mismatch.

[1m [36m..... [39m [22m